

FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text

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by

Muhib Al Ahad

Roll: 22201207

Faiza Haque Shoily

Roll: 22201183

Tanjuma Noor

Roll: 22201210

Supervised by:

Shammi Akhtar

Assistant Professor

Co-supervised by:

Md Shafayatul Haque

Lecturer



Department of Computer Science & Engineering
University of Asia Pacific

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Declaration

We, hereby, declare that the work presented in this Thesis is the outcome of the investigation performed by us under the supervision of Ms. Shammi Akhtar, Assistant Professor, Department of Computer Science, University of Asia Pacific. We also declare that no part of this Thesis and thereof has been or is being submitted elsewhere for the award of any degree or Diploma.

Countersigned

(Ms. Shammi
Akhtar)
Supervisor

Signature

.....

Faiza Haque Shoily

.....

Muhib Al Ahad

.....

Tanjuma Noor

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Abstract

Sharing medical data across hospitals could unlock huge advances in healthcare AI, but it comes with a real problem: patient records are sensitive, and no institution can simply hand over raw data to build a shared model. This tension between collaboration and privacy is exactly what motivated this work. To navigate it, we turned to Federated Learning (FL), an approach that lets multiple hospitals train a shared model together while each one keeps its own patient data locally, never sending it anywhere.

We applied this idea to Bangla medical text, a space that has been largely overlooked. While English has been the focus of most medical NLP research, with vast datasets and pretrained models readily available, Bangla has not received the same attention, despite being spoken by hundreds of millions of people. This gap means that many of the tools clinicians and researchers rely on in English simply don't exist for Bangla, and patients whose records are written in Bangla are effectively left out of these advances. Motivated by this gap, we developed FedMed-BN, a privacy-preserving federated learning framework built specifically for Bangla medical text.

We see this work as more than a technical exercise. By bringing together healthcare NLP and federated learning in a language that has been underserved, FedMed-BN aims to be a genuine step toward more inclusive medical AI, one that could eventually support better prescription analysis, disease classification, and clinical documentation tools for Bangla-speaking communities, without asking hospitals to compromise on patient privacy to get there.

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Chapter 1

Introduction

1.1 Background and Motivation

In Bangladesh, the swift digitalization of healthcare has resulted in an explosion of unstructured clinical text in Bangla, including prescriptions, symptom descriptions, discharge summaries and electronic health record (EHR) notes. Despite this demographic scale, Bangla is significantly underrepresented in the medical NLP literature, a disparity that remains even when compared to languages with much smaller speaker populations but more established clinical NLP infrastructures [20]. This underrepresentation is further aggravated by a structural constraint inherent to the Bangladeshi healthcare system. Clinical data is dispersed across hospitals, diagnostic chains and specialist clinics which operate as isolated institutional silos due to confidentiality obligations, patient privacy considerations and the absence of a unified national health data policy. Consequently, centralized aggregation of such data for model training is rendered both ethically untenable and practically infeasible, rendering conventional deep learning pipelines - which presuppose centralized data access-fundamentally incompatible with the operational realities of Bangladeshi clinical informatics. This study is motivated by the imperative to address this dual gap by developing high-performance Bangla clinical NLP methods that obviate the need for raw patient data to leave its institutional source [9].

1.2 Problem Statement

The primary concern that this work addresses is the simultaneous occurrence of three connected issues, namely, the lack of effective Bangla-specific clinical language models, the inability to centralize sensitive patient data across institutions, and the vulnerabil-

ity of standard decentralized training approaches to adversarial reconstruction attacks, known as gradient leakage. While current Federated Learning (FL) frameworks are designed to avoid sharing raw data, advanced inference attacks have been shown to be able to mathematically invert private text sequences from transmitted model weight updates [15]. Thus, network isolation alone is insufficient to ensure the confidentiality of patients. Taken together, these three issues define a problem that no existing approach fully resolves: an effective clinical NLP solution for Bangla must simultaneously (i) achieve competitive performance on a morphologically rich, low-resource language, (ii) operate under the institutional, legal, and ethical constraints that prevent centralized data pooling in Bangladeshi healthcare, and (iii) provide a formal, quantifiable privacy guarantee that holds even against adversaries with access to the shared model updates themselves [21]. This research is positioned to address this composite problem directly, rather than treating any one of its dimensions in isolation.

1.3 Objectives of the Project

To address these overlapping challenges, this research pursues the following objectives. First, it establishes a Federated Learning framework specifically optimized for unstructured Bangla medical text, simulating a realistic multi-institutional hospital environment using distributed data partitions. Second, the system utilizes a technique referred to as differentially private federated averaging (DP-FedAvg), where calibrated Gaussian noise is added to the gradients of the local models before their aggregates are performed by the server. It fine-tunes a pre-trained Bangla transformer model—specifically csebuetnlp/banglabert—for Bangla medical named entity recognition (NER), identifying entities such as Disease, Medicine, Organ, Hormone, Pharmacological Class, and Common Medical Terms. Fourth, it evaluates model convergence and stability under Non-IID data distributions, which arise naturally when different institutions exhibit different diagnostic patterns and documentation styles.

1.4 Research Questions Guiding the Investigation

This study is guided by the following research questions (RQs), organized around five core dimensions: correct application of subject-specific concepts, logical consistency, validity

and justification, innovation and contribution, and overall quality of the proposed system.

RQ Group 1: Correct Application of Subject-Specific Concepts

RQ1.1 Does FedMed-BN correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than merely simulating distributed training without true data isolation?

RQ1.2 Is BanglaBERT an appropriate and correctly utilized pretrained model for processing Bangla medical text?

RQ1.3 Is the FedAvg aggregation mechanism implemented and explained correctly, particularly with respect to weighted averaging by client data size?

RQ Group 2: Logical Consistency

RQ2.1 Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text?

RQ2.2 Is the use of non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics or devices would differ from one another?

RQ2.3 Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project’s privacy-preservation claims?

RQ2.4 Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)?

RQ Group 3: Validity and Justification

RQ3.1 Is there sufficient justification for choosing BanglaBERT over alternatives such as mBERT or XLM-RoBERTa?

RQ3.2 Is the privacy-preservation claim adequately justified given that only architectural separation (without formal differential privacy or encryption) is used?

RQ3.3 Is the simulated client partitioning method adequately explained and justified?

RQ3.4 Are the performance metrics (accuracy, F1-score, precision, recall) appropriate and justified for evaluating Bangla medical NER specifically?

RQ Group 4: Innovation and Contribution

RQ4.1 Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (e.g., MIMIC-III, i2b2) or generic text classification benchmarks?

RQ4.2 Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource, non-English languages)?

RQ4.3 Is there a clear, realistic practical application for this work within the Bangladeshi healthcare context?

RQ Group 5: Overall Quality

RQ5.1 Does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem?

RQ5.2 Are the project’s stated limitations and future work realistic and proportionate to its current stage of development?

RQ5.3 Would this work constitute a meaningful and publishable contribution to federated NLP for low-resource medical text?

1.5 Scope of the Work

The work is restricted to Bangla medical named entity recognition (NER), performed on Bangla-language clinical text drawn from the Bangla-MedER and BanglaBioMed corpora. One pre-trained Bangla transformer backbone, csebuetnlp/banglabert, is fine-tuned within the federated setting; the design and pre-training of these base models is outside the scope of this work. The multi-institutional environment is simulated in software using five distributed data partitions that emulate realistic, non-IID hospital data distributions; the study does not involve deployment within actual hospital infrastructure or live patient data. Privacy protection is provided through DP-FedAvg, with the privacy budget formalized using the (ϵ, δ) -differential privacy definition; other privacy-enhancing techniques, such as secure multi-party computation or homomorphic encryption, are noted in the literature but are not implemented here. Finally, the study is limited to the

textual modality of clinical data; medical imaging, structured tabular EHR fields, and speech-based clinical data are not considered.

1.6 Expected Outcomes

This research is expected to produce an end-to-end Federated Learning pipeline tailored to Bangla medical text, in which differential privacy is incorporated as a first-class design component rather than an afterthought. Unlike the overwhelming majority of existing Bangla clinical NLP studies, which require all training data to be pooled in a single location, the proposed framework is expected to demonstrate that collaborative model training across five simulated hospital nodes is achievable without any raw text ever leaving its local environment. The work is also expected to yield a structured experimental benchmark built on established Bangla clinical corpora, including the Bangla-MedER and BanglaBioMed datasets, partitioned under realistic Non-IID constraints to simulate genuine institutional heterogeneity. A further expected outcome is a quantifiable and auditable measure of patient data protection, expressed through the (ϵ, δ) -differential privacy budget, that makes the trade-off between privacy and utility explicit and adjustable. Collectively, these outcomes are intended to establish a replicable blueprint for privacy-preserving medical NLP in other low-resource linguistic contexts, with particular relevance for South and Southeast Asian healthcare systems facing similar data fragmentation and linguistic diversity challenges.

1.7 Impacts of the Project

Beyond its technical contributions, this project carries implications that extend into societal, health and legal, and environmental domains. The following subsections outline these broader impacts.

1.7.1 Impact on Societal and Cultural Issues

By targeting Bangla—a language spoken natively by over 170 million people in Bangladesh and more than 230 million people worldwide—this project works toward closing a long-standing technological gap that has left Bangla-speaking patients underserved by modern clinical AI tools [11]. Improved Bangla clinical NLP capability can, over time, support

better documentation, retrieval, and decision-support tools for clinicians who work primarily in Bangla, rather than forcing reliance on English-centric systems [24]. This has a cultural dimension as well: it affirms that native-language clinical documentation does not need to be a barrier to adopting advanced AI tools, and it contributes to the broader movement toward linguistic inclusion in NLP research and digital health infrastructure [30, 26].

1.7.2 Impact on Health, Safety, and Legal Issues

The federated, privacy-preserving design of this work is directly motivated by patient safety and legal considerations. By ensuring that raw patient text never leaves its institutional source, and by formalizing a quantifiable (ϵ, δ) privacy budget, the framework is intended to reduce the risk of patient re-identification or data exposure relative to conventional centralized training pipelines [2]. This is particularly relevant in Bangladesh, where institutional confidentiality obligations and patient privacy ethics already constrain data sharing, even in the absence of a single comprehensive national health data policy. At the same time, any system that processes drug names, dosages, and disease identifiers carries health and safety responsibilities. The NER and classification capabilities developed in this work are intended as decision-support aids that can assist clinicians and researchers in structuring unstructured notes, not as an autonomous diagnostic authority; outputs would require clinical validation before influencing patient care decisions [25, 3]. Care has also been taken to evaluate model performance under realistic, non-IID conditions, so that reported accuracy is not artificially inflated by favorable, idealized data distributions.

1.7.3 Impact on Environment and Sustainability Issues

Training and fine-tuning transformer-based language models carries a computational, and therefore energy, cost, and the federated setting used here introduces additional communication overhead from repeated rounds of model-update exchange between simulated hospital nodes and the central server. This project mitigates some of this cost by fine-tuning existing pre-trained Bangla transformer backbones rather than training large models from scratch, and by working with modestly sized clinical corpora rather than web-scale data.

More broadly, the federated approach itself has a sustainability rationale: because raw data never has to be duplicated, transferred, and stored centrally, this design avoids the storage and transmission overhead—and corresponding energy footprint—of large-scale centralized data pooling [13, 8]. In this sense, the privacy-preserving architecture investigated in this work also aligns with the goal of building clinical AI infrastructure that scales sustainably across multiple institutions, rather than depending on ever-larger centralized repositories of sensitive data.

1.8 Report Organization

The remainder of this report is organized as follows:

- **Chapter 1** introduces the background and motivation, problem statement, objectives, research questions, scope, expected outcomes, and broader impacts of the project.
- **Chapter 2** presents the background theory of Federated Learning, Differential Privacy, and transformer-based Bangla NLP, followed by a detailed literature review, comparative analysis of related works, and the identified research gap.
- **Chapter 3** covers the requirement analysis (including stakeholder questionnaires and functional/non-functional requirements), the system architecture, detailed design diagrams, and the design alternatives considered for FedMed-BN.
- **Chapter 4** describes the methodology and implementation, including the tools and libraries used, data collection and preprocessing, model development, and system implementation details.
- **Chapter 5** presents the experimental setup, performance metrics, result analysis, and limitations of the evaluation.
- **Chapter 6** outlines the project planning and budget, including the project timeline and cost estimation.
- **Chapter 7** maps the project to relevant engineering standards and accreditation requirements, covering Complex Engineering Problem (CEP) and Complex Engineering Activity (CEA) attributes.

- **Chapter 8** concludes the report with a summary of key findings and contributions, limitations of the study, and recommendations for future work.

Chapter 2

Background and Literature Review

This chapter presents the theoretical foundation and reviews related research relevant to our project on Privacy-Preserving Federated Learning for Bangla Medical Text. It begins with essential concepts and definitions in the Preliminaries, followed by a detailed survey of existing systems. The chapter then compares these methods, highlights their limitations, and identifies research gaps addressed by our proposed system.

2.1 Preliminaries

In this chapter we outline the basic ideas that will allow us to use FedMed-BN.

Federated Learning (FL)

Federated Learning was introduced by Google [14] for supporting the data privacy-sensitive character of distributed learning. It allows multiple clients to collaborate in training a single model of shared knowledge, without having to share any client’s raw data. Instead, each party trains the model individually on their own data and shares only the changes they made to the model since the last round — called “updates” or “gradients” — with the other parties. The changes are aggregated by the central server, which then returns the updated global model to all parties. This process is repeated for multiple communication rounds.

Formally, FedAvg minimizes the global objective:

$$F(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w), \quad F_k(w) = \frac{1}{n_k} \sum_{i \in \mathcal{P}_k} f_i(w) \quad (2.1)$$

where K is the number of participating clients, n_k is the number of data samples held by

client k , $n = \sum_k n_k$ is the total number of samples across all clients, $\mathcal{P}_k$ denotes the set of data indices on client k , and $f_i(w)$ is the loss of the model parameters w on sample i .

Differential Privacy (DP)

Differential Privacy guarantees that no matter how much information is contained in an algorithm’s output, it cannot be determined whether any individual has contributed, or whether any individual’s data was used. To achieve differential privacy in federated learning, gradients are clipped and Gaussian noise is added before sharing model updates with the server. The DP-FedAvg algorithm calculates:

$$\bar{\Delta}_t = \frac{1}{L} \left(\sum_{k \in S_t} \text{clip}(\Delta_k^t, C) + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \right), \quad w^{t+1} = w^t + \bar{\Delta}_t \quad (2.2)$$

where S_t is the set of L clients sampled at round t , Δ_k^t is the model update from client k , C is the clipping norm bound, and σ is the noise multiplier controlling the strength of the Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ added to the aggregated update.

Transformer Models and BanglaBERT

BERT (Bidirectional Encoder Representations from Transformers) is a type of pre-trained language model that uses learning to acquire rich contextual representations of text. BanglaBERT is a BERT based pre-trained model that has been trained using large amounts of Bangladeshi written text. As such, it can capture the structural aspects of the Bangla language and then fine tune the model for specific downstream tasks by attaching task specific heads; for example, we attach a token classification head to recognize medical entities such as Disease, Medicine, and Organ in Bangla clinical text.

Non-IID Data in Federated Learning

In a real-world federated learning environment, the data from one client to the next is rarely independent and identically distributed; it is non-IID. Take hospitals for example: they will have their own mix of patients and diseases as well as distinct ways of documenting them. Such heterogeneity poses a difficulty for federated training, which is precisely the sort of issue our system is built to handle with the DP-FedAvg framework.

2.2 Related Works / Existing Systems

Previous studies in federated learning applied to medical NLP are diverse. We review the existing research in three categories: federated NLP systems, privacy-preserving federated learning, and Bangla medical NLP.

Federated Learning for Medical Text Classification

Chen et al. [12] present FLCMC, a federated learning approach to Chinese medicinal text categorization. FedPA and FedPAP - perturbed federated algorithms - were introduced and evaluated on the IMCS-V2 dialogue dataset across simulated non-IID clients. FedPAP reached a peak accuracy of 0.79, indicating that federated techniques are able to perform on par with centralized baselines in medical text. However, their paper is entirely on Chinese text and no differential privacy is used.

Huang et al. [18] proposed a federated ICD-10 classification system with PubMedBERT across three Taiwanese hospitals as distinct data silos. The federated model achieved an F1-score of 0.812, a 3.11% improvement over the single-hospital baseline. The study shows the real-world utility of federated BERT models in the clinic, but the authors admit that differential privacy had not been included or integrated at the time of writing.

Liu et al. [28] addressed communication efficiency in federated text classification with FedInitPrune, a parameter pruning algorithm. When applied to a federated TextCNN setup, this resulted in a 74.26% reduction in communication parameters and a less than 2% decrease in accuracy, representing a significant real-world benefit for resource-limited hospital settings.

Agrawal et al. [27] performed an extensive systematic literature review of federated learning for NLP tasks, covering FedAvg, FedProx, FedAdam, LSTM, and BERT-based methodologies across different datasets. The review highlights key issues such as non-IID data heterogeneity, communication restrictions, and the inherent privacy-performance trade-off.

Hasan et al. [19, 2] classified medical prescriptions using an OCR pipeline (EAST detector with Tesseract) combined with rule-based NLP tagging, reporting 93–95% accuracy on white-prescription detection; although not federated, the work illustrates the domain-specific preprocessing challenges typical of clinical text.

Dligach and Miller [9] proposed a two-stage federated NLP pipeline that first learns

patient representations with a Deep Averaging Network and then performs federated phenotyping, achieving an F1-score statistically equivalent to a centralized baseline on MIMIC-III and the i2b2 Obesity Challenge dataset, without any differential privacy mechanism.

Zhao et al. [35] explored personalized federated text classification through gradient-free prompt tuning with a genetic algorithm, reducing communication cost by roughly 1000× relative to gradient-based FL while remaining competitive in accuracy, though at the cost of heavier local (edge) computation. In the relation-extraction setting, [30] introduced FedED, a federated ensemble-distillation approach for medical relation extraction that avoids sharing raw gradients between clients, further demonstrating that federated strategies can be adapted across different medical NLP sub-tasks beyond classification.

Privacy-Preserving Federated Learning

Lyu et al. [31] introduced a privacy-preserving federated learning approach integrating Local Differential Privacy (LDP), bitwise quantization, and feature hashing for natural language processing on edge nodes. On sentiment analysis and rating prediction tasks, they showed that this was possible without costly embeddings, although how well the approach performs on challenging medical NLP tasks has not been validated.

Ahmed et al. [37] proposed a privacy-preserving edge-aggregated federated model using Shamir’s Secret Sharing for healthcare severity classification. The proposed hierarchical aggregation with dynamic edge thresholding showed promising privacy protections on the MIMIC-III dataset, but the authors acknowledge that the method does not currently extend to medical text or NLP tasks.

Gawali et al. [4] performed federated deep learning for lung cancer segmentation across 12 hospitals in 8 countries using Personal Health Train infrastructure. Their research shows that cross-institutional federated learning without data leaving a hospital is practically possible, but communication lags due to slow hospital connections remain an issue.

Rahman et al. [14] introduced FedHealth-NLP, integrating FedBERT with differential privacy and a knowledge graph neural network for community health literacy. Their method obtained an accuracy of 91.3% across four community health datasets, 4.4% better than FedAvg+BERT and, at the same time, 42% less communication cost.

In-Depth FL Evaluation for Biomedical NLP

Shen et al. [29] systematically evaluated FL across eight biomedical NLP corpora, including NER and relation extraction, observing that FL models consistently outperformed single-client training and notably outperformed LLMs (GPT-4, PaLM 2) in the few-shot prompting scenario as well. Importantly, they identified federated learning with differential privacy as a research opportunity but did not implement it themselves.

Gao et al. [10] introduced FedCliMask, an ontology-guided semantic masking framework using UMLS concept hierarchies for clinical NLP. They implemented a federated training approach in which four levels of semantic masking using ClinicalBERT on synthetic hospital notes indicated the possibility of meaningful privacy–utility tradeoffs. However, they did not implement their full federated training pipeline.

General Medical Text Classification and NLP Techniques

Beyond federated and privacy-preserving settings, a broader body of work has studied medical text classification in the (non-federated) centralized setting, and this line of research informs the modeling choices used in FedMed-BN. [20] compared several deep learning architectures for classifying unstructured medical notes under varying degrees of class imbalance, a challenge that is directly relevant to non-IID federated clients. [11] conducted a systematic review of NLP techniques for biomedical document classification, cataloguing feature representations ranging from bag-of-words to transformer embeddings. [25] applied linguistic-feature-driven supervised deep learning to unstructured medical text classification, while [3] presented an end-to-end text classification pipeline built on state-of-the-art NLP models. Closer to the prescription-processing angle of [19], [26] performed medical text extraction and classification directly from prescription images, combining OCR with downstream NLP classification. Taken together, these centralized studies establish strong non-federated baselines but, like the federated works reviewed above, none of them jointly address federation, differential privacy, and low-resource languages.

Bangla Medical NLP

Islam et al. [7] presented BanglaBioMed, the first Bangla biomedical NER corpus, from BioNLP@ACL 2022. BanglaBERT obtained the best F1-score of 81.33% on this corpus,

covering anatomy, chemicals/drugs, disease/symptom, and medical procedure entities. The proposed corpus and the performance achieved by BanglaBERT both indicate the suitability of transformer-based NLP approaches for Bangla biomedical text.

Dhar et al. [38] introduced a hybrid internal feature approach, considering word length and English-equivalent terms as features for categorizing Bangla medical text documents. The proposed SGD classifier obtained 97.75% accuracy in binary medical vs. non-medical classification. This is one of the earliest works in Bangla medical text categorization.

2.3 Comparative Analysis of Related Works

Table 2.1 summarizes and compares the reviewed works across dataset, method, focus, performance, and key limitations.

Table 2.1: Summary of Related Work in Federated Learning for Medical/Biomedical NLP

Research	Year	Dataset	Method / Model	Main Focus	Performance / Strength	Limitation
Chen et al. (FLCMC) [12]	2024	IMCS-V2 (4,116 doctor-patient dialogues, 10 pediatric diseases); Synthetic non-IID datasets	FLCMC; FedPA (Perturbed Adam Federated Learning)	Federated medical text classification for Chinese medicinal dialogue	FedPAP: 0.79 accuracy, 0.48 loss; Competitive with centralized baselines	No differential privacy; Chinese only; text-only; large model deployment impractical on edge clients
Hasan et al. (Medical Prescription) [19]	2019	~800,000 text rows from medical prescriptions; White Prescription dataset (75/25 split)	OCR (EAST + Tesseract); Supervised ML Classifier; Damerau-Levenshtein spelling correction; Rule-based NLP tagging	NLP-based tagging of medical prescriptions using OCR and rule-based tagging	93–95% accuracy in white prescription detection; Only 0.5% required manual intervention	Manual rule creation time-consuming; 30% initially unclassifiable; Limited to prescription domain only
Dligach et al. (Two-stage Fed. Phenotyping) [9]	2019	MIMIC-III (58,000+ admissions); i2b2 Obesity Challenge (1,237 discharge summaries)	Two-stage Federated NLP: Deep Averaging Network (DAN) for patient representation + federated phenotyping classifier	Federated patient phenotyping and representation learning across hospitals without data sharing	Fully federated F1 = 0.724; Statistically equivalent to centralized approach	Impractical for rare conditions; Cannot directly compare to i2b2 benchmarks; Data heterogeneity across hospitals

Zhao et al. (Gradient-Free Prompt Tuning) [35]	2024	SST-2, AG's News, TREC; split into 10–100 non-IID silos	RoBERTa-large + Genetic Algorithm (GA) for personalized federated local prompt optimization	Personalized federated text classification with gradient-free prompt tuning	Reduced communication cost by 1000x; Competitive accuracy vs gradient-based FL baselines	GA computationally intensive on edge devices; Heavily dependent on initial prompt templates
Islam et al. (BanglaBioMed) [27]	2022	BanglaBioMed: 818 sentences, 11,196 tokens; 4 entity types; 70/10/20 train/-val/test split	BanglaBERT, mBERT, XLM-RoBERTa fine-tuned for Bangla biomedical NER	First Bangla biomedical NER corpus and baseline evaluation	BanglaBERT F1 = 81.33% (best); Outperforms mBERT and XLM-RoBERTa	No federated learning; No privacy mechanism; Small dataset (818 sentences); Entity class imbalance
Huang et al. (ICD-10 FL) [31]	2023	Multicenter EMRs from 3 Taiwanese hospitals (NTUH, FEMH, VGHTPE) as 3 silos	Federated PubMedBERT, BioBERT, ClinicalBERT with Label Attention	Federated ICD-10 classification across real hospital silos	Federated PubMedBERT F1 = 0.812 (3.11% improvement over single-hospital baseline)	No differential privacy; Non-IID challenges from varying hospital styles; High local compute required
Liu et al. (FedInit-Prune) [37]	2022	Fudan Chinese Text Classification Dataset; THUCNews	FedW2V + Federated TextCNN + FedInitPrune (parameter pruning algorithm)	Communication-efficient federated text classification via parameter pruning	74.26% reduction in communication parameters with less than 2% accuracy drop vs FedAvg	No privacy mechanism; Pruning ratio manually chosen; Chinese only; Does not address model heterogeneity
Ahmed et al. (FL Healthcare Big Data) [4]	2025	TB Chest X-ray images (Kaggle); 7,000 images; 10 simulated client nodes	ResNet (primary); DenseNet; SqueezeNet; FedAvg with SMOTE for class balancing	Privacy-preserving federated deep learning for distributed healthcare TB diagnosis	ResNet: 96.7% accuracy, 97.4% F1-score; ROC AUC = 0.97	Single imaging modality only; SMOTE may introduce artifacts; No DP; No NLP component
Agrawal et al. (FL-NLP SLR) [15]	2024	Gboard, Stack Overflow, Shakespeare, UCI Spam, 20 News, IMDB, Sentiment140, CoLA	Survey of FedAvg, FedProx, FedAdam, LSTM, RNN, GPT-2, BERT, NMF	Systematic review of FL-based NLP across prediction, classification, and generation	92% spam classification; 76% ternary sentiment; 51.49% search suggestions	Survey only; Non-English NLP underexplored; Privacy-performance trade-off unresolved
Lyu et al. (LDP-FL NLP) [38]	2023	IMDB Movie Reviews (sentiment); MovieLens with keywords (rating prediction)	LDP + Bitwise Quantization + Feature Hashing; Privacy-preserving FL for NLP on edge devices without raw data collection	Feasible privacy-preserving FL on edge without resource-heavy embeddings	Competitive accuracy without resource-heavy embeddings	Rolling-hash loses semantic richness; LDP noise degrades accuracy; Not validated on medical NLP

Ahmed et al. (Edge-FL Healthcare) [21]	2025	Synthetic health-care data + real MIMIC-III ICU dataset	Hierarchical FL with Shamir’s Secret Sharing (SMPC) + dynamic edge thresholding	Privacy-preserving severity classification in healthcare via secure edge aggregation	Strong privacy via SMPC; Evaluated with accuracy, F1, confusion matrix across 5 runs	Structured/tabular data only — no NLP; No fairness or explainability
Gawali et al. (PHT / FedCT) [33]	2025	Lung cancer chest CT scans from 12 hospitals, 8 countries, 2,078 patients	FedAvg + Personal Health Train (PHT) / Vantage6 with Secure Aggregation Server	Real-world cross-institutional FL for lung cancer GTV segmentation	Feasibility across 12 real hospitals; No patient data left any institution during training	Imaging only — not NLP; No formal DP; Bottlenecked by slowest hospital
Rahman et al. (FedHealth-NLP) [24]	2025	HealthLit-Urban (45,230 docs); RuralHealth-USA (23,456 docs); MultiLingual-Health (67,890 docs); CommunityEngagement (31,245 docs)	FedBERT + DP gradient aggregate + PKGNN (Knowledge Graph NN) cross-modal attention fusion	Federated NLP with DP for community health literacy enhancement	91.3% accuracy, F1 = 0.898; +4.4% over pFedMe; 42% communication reduction	Simulated nodes — not real deployment; DP reduces accuracy by 0.8%; English only
Shen et al. (FL Biomedical NLP) [22]	2024	8 biomedical NLP corpora: BC2GM, BC4CHEMD, JNLPBA, NCBI-disease (NER); EUADR, GAD, n2c2 (RE)	FedAvg, FedProx; BERT, BioBERT, BlueBERT, Bio_ClinicalBERT, GPT-2, BiLSTM-CRF	Systematic evaluation of FL on biomedical NER and relation extraction	FL > single-client; FedProx F1 = 0.994 under non-IID; FL significantly outperforms GPT-4/PaLM 2 few-shot	Simulated splits — not real FL; Horizontal FL only; No DP or homomorphic encryption integrated
Patel et al. (FedCli-Mask) [10]	2025	12,996 synthetic clinical notes (Synthea) across 6 simulated US hospital types	UMLS ontology-guided 4-level semantic masking + ClinicalBERT NER + proposed CA-FedAvg	Context-aware FL with ontology-guided masking for privacy-utility balance in clinical NLP	Level 2 masking identified as optimal privacy-utility balance; 6 hospital profiles successfully differentiated	CA-FedAvg proposed but not implemented; Synthetic data only; English only; No formal DP guarantees

Each of the reviewed techniques has its own set of useful attributes as well as significant disadvantages. Prescription-based and rule-based NLP systems are very accurate but can only be applied to their specific domain. Federated phenotyping approaches show that FL is comparable to centralized performance for clinical text, but it does not come with privacy-enhancing mechanisms other than data locality. Parallel techniques such as FedInitPrune can dramatically cut down on transmission expenses, but only in a single language and without any privacy protection whatsoever. Medical imaging FL systems

like the TB diagnosis framework and the PHT lung cancer study prove feasibility in a real-world setting, but cannot be applied to text-based NLP tasks across hospital networks. We also evaluate privacy-preserving frameworks, such as LDP-FL and FedHealth-NLP, in which differential privacy is combined with federated NLP and evaluated on English datasets. The extensive biomedical NLP evaluation by Shen et al. also demonstrates that FL consistently outperforms single-client training and LLMs under few-shot prompting, and explicitly notes that the integration of DP is an open challenge. On the Bangla side, BanglaBioMed and Dhar et al. establish strong baselines for Bangla medical NLP, but neither is equipped with federated learning or privacy protection, making them difficult to deploy in real multi-institutional settings. Overall, these methods represent good individual approaches, but none cover all the needs of differential privacy, federated training, and low-resource Bangla medical NLP, which is what FedMed-BN addresses.

2.4 Research Gap / Limitations of Existing Methods

The literature review clearly shows a notable research gap. Although federated learning has been used for medical NLP in English and Chinese, federated learning and differential privacy have not been combined for Bangla medical text in particular. We were motivated by the following gaps to develop FedMed-BN:

1. **Language gap:** All the FL medical NLP systems we reviewed work on English and Chinese. Bangla, despite having 230 million speakers, does not have any federated medical NLP framework.
2. **Privacy gap:** Most federated medical NLP systems do not integrate differential privacy, which makes them susceptible to gradient leakage attacks.
3. **Domain gap:** Existing Bangla medical NLP works operate in centralized settings and have no privacy protection, making them unfeasible for real multi-hospital deployment.
4. **Resource gap:** There is currently no system that takes into consideration the federated learning challenges of a low-resource language like Bangla, specifically in the medical domain.

FedMed-BN tackles all four gaps by integrating DP-FedAvg with BanglaBERT for privacy-preserving Bangla medical named entity recognition.

2.5 Contribution

This work contributes to the research gap identified in Section 2.4 as follows:

1. **First ever federated learning approach for Bangla named entity recognition.** Although federated learning has been well studied in English (MIMIC-III, i2b2) and Chinese (IMCS-V2), no system has been developed to use federated learning for Bangla clinical and biomedical texts. FedMed-BN is placed as the first attempt to remove this language barrier of Bangla medical NER.
2. **Formal, quantifiable privacy guarantee, not just architectural separation.** The majority of federated medical NLP systems discussed in Section 2.2 (such as Chen et al.
3. **Linking Bangla medical NLP to a true multi-institutional context.** The current Biomedical NER Bangla (BanglaBioMed
4. **A federated benchmark for Bangla clinical health care that is reusable.** This work proposes a benchmark setup, which is built on realistic non-IID client shards of Bangla-MedER and BanglaBioMed and recognize the entity types like Disease, Medicine, Organ, Hormone, Pharmacological Class, and Common Medical Terms, so that future Bangla clinical NLP research can build upon this setup without designing another federated setup from scratch.

Combined, these contributions address the four gaps (language, privacy, domain, and resource) identified in Section 2.4, making FedMed-BN different from all the surveyed systems in this chapter.

2.6 Summary

In this chapter, we reviewed the theoretical foundations of federated learning, differential privacy, and transformer-based NLP. Significant achievements in medical text processing

were identified through a literature survey of federated NLP architectures. The comparison of existing systems revealed a gap: no existing system combines federated learning with differential privacy for Bangla medical NLP. The proposed FedMed-BN framework is directly motivated by this gap, which will be explained in the following chapters.

Chapter 3

System Analysis and Design

3.1 Requirement Analysis

3.1.1 Questionnaires

To construct a model that meets the expert's needs, we held several interviews to evaluate the functional and non-functional requirements of our proposed model. The mentionable experts that we interviewed are:

Name	Occupation	Contact Info
Shayela Begum	Physician	shayelabegum69@gmail.com
Zakir Hossain	Surgeon	zplastic61@gmail.com
Nasima Kamal	Doctor	nasimakamal1972@gmail.com

Table 3.1: Expert Evaluators Consulted for FedMed-BN Requirement Analysis

Below is the table of questions and the generalized answers received from the conducted interviews:

In future, we plan to conduct additional expert interviews as needed during development to ensure the model meets user needs.

Question	Feedback
Does the project correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than simply simulating distributed training without true data isolation?	Yes, local training with no raw data sharing is essential for patient privacy.
Is BanglaBERT an appropriate and correctly used pretrained model for Bangla medical text?	Absolutely, BanglaBERT is pre-trained on a large Bangla corpus, making it well-suited for understanding Bangla medical terminology and clinical text.
Is the FedAvg aggregation mechanism implemented and explained correctly (e.g., weighted averaging by client data size)?	Yes, weighting by data size makes sense when hospitals have very different patient volumes.
Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text?	Absolutely, centralizing patient records would raise serious privacy and compliance concerns.
Is the use of non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other?	Yes, every clinic sees different patients and writes notes differently, so this is realistic.
Is there sufficient justification for choosing BanglaBERT over alternatives such as mBERT or XLM-RoBERTa?	Yes, but a direct comparison with mBERT and XLM-RoBERTa would strengthen the justification.
Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used?	Partially, architectural separation helps, but formal privacy guarantees would be needed for real deployment.
Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks?	Yes, Bangla medical NLP is underexplored, so this fills a real need.
Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context?	Yes, this could genuinely help hospitals and clinics collaborate without sharing sensitive patient data.

Table 3.2: Interview Questions and Generalized Expert Responses

3.1.2 Functional and Non-Functional Requirements

The functional and non-functional requirements were distilled from a review of closely related literature on federated learning for medical/biomedical NLP, insights from the replicated baseline experiments, and questionnaires from expert interviews, with each requirement converted into a concise, testable statement with clear acceptance criteria.

Table 3.3: Functional and Non-Functional Requirements

ID	Description	Collected From
FR-1	The system shall train the BanglishBERT model locally on each simulated client’s Bangla-English medical text without transmitting raw patient data to the central server.	Expert-1, Expert-2
FR-2	The system shall aggregate client model updates at the central server using FedAvg, weighted by each client’s local data size.	Expert-1, Expert-2, Expert-3
FR-3	The system shall accept Bangla medical text as input and identify medical entities such as Disease, Medicine, Organ, Hormone, Pharmacological Class, and Common Medical Terms.	Expert-1, Expert-2, Expert-3
FR-4	The system shall support partitioning the dataset into non-IID client shards to simulate realistic variation across different clinics or devices.	Expert-1, Expert-3
FR-5	The system shall compute accuracy and perplexity metrics separately for the federated model and a centralized baseline to enable direct comparison.	Expert-2, Expert-3
NFR-1	The system shall ensure raw patient text never leaves the local client during training; only model weight updates shall be transmitted.	Expert-1, Expert-2
NFR-2	The system shall remain compatible with alternative pretrained models (e.g., mBERT, XLM-RoBERTa) to allow substitution and comparison against BanglaBERT.	Expert-3
NFR-3	The system shall minimize communication overhead between clients and the central server across training rounds.	Expert-2, Expert-3
NFR-4	The system shall ensure all experiments are reproducible through fixed random seeds and clearly documented hyperparameters.	Expert-1, Expert-3

3.1.3 Stakeholder Engagement Evidence

The research team engaged with multiple stakeholders to validate requirements and ensure methodological rigor.

Key Insights

The expert consultations highlighted several important recommendations that guided the development of the proposed framework. The key insights are summarized below:

- **Shayela Begum:** Through questionnaires, she gave feedback on the clinical relevance of the Bangla-English medical text samples and recommended incorporating real-world prescription phrasing to improve the practical applicability of the federated model.
- **Zakir Hossain:** Focused on the privacy implications of handling patient text across distributed clients, and emphasized the importance of clearly documenting what data leaves the local client versus what remains on-device.
- **Nasima Kamal:** Reviewed the proposed framework and suggested prioritizing evaluation against realistic non-IID clinic data distributions, since real Bangladeshi healthcare settings vary significantly in patient volume and note-taking style.

Table 3.4: Expert Consultation Evidence

Date	Expert Name	Affiliation	Purpose
[28-05-2026]	Shayela Begum	Physician	Guidance on clinical relevance of Bangla-English medical text samples and prescription phrasing.
[04-06-2026]	Zakir Hossain	Surgeon	Consultation on privacy implications and client-side versus server-side data handling.
[24-07-2026]	Nasima Kamal	Doctor	Expert review focusing on non-IID data realism and evaluation design.

Evidence and Documentation

Supporting evidence for requirement identification, including stakeholder consultation summaries, is provided in Appendix D.

3.2 System Overview / Architecture

This section describes the overall architecture of FedMed-BN. To accommodate several simulated hospital clients to train a shared BanglaBERT-based medical text classifier without sharing any raw patient information, the system has been designed as a data privacy-preserving federated learning pipeline.

3.2.1 High-Level Architecture

At a high level, FedMed-BN consists of five simulated hospital clients (Hospital A through E) and a central aggregation server. Each client has its own partition of the Bangla medical text dataset, which is non-overlapping. Throughout the round of communications, clients update their local model using their own data and add Gaussian noise and gradient clipping to protect their information, then return only the differentially private updates to the central server. These updates are compiled using FedAvg on the server, and then the aggregated global model is sent back to all the clients. This continues for T communication rounds.

The high-level block diagram of FedMed-BN is presented in Fig. 3.1.

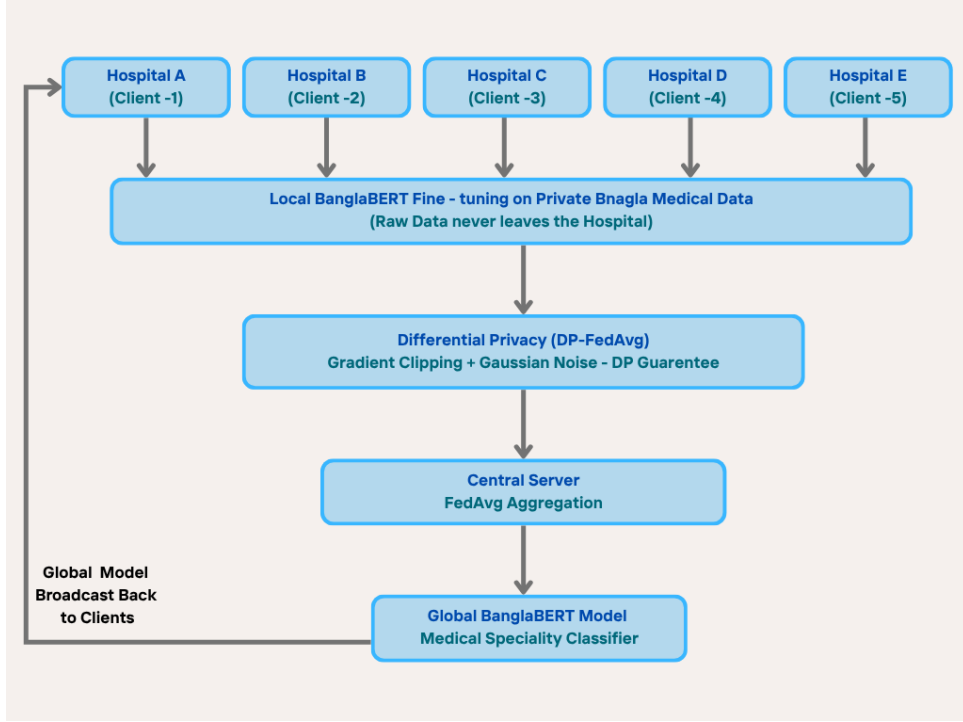


Figure 3.1: High-Level Architecture of FedMed-BN

3.2.2 Detailed Design Diagrams

The system is described in more detail in the following diagrams. The Differential Privacy mechanism is applied at each client before sending the model updates to the server, as shown in Figure 3.2. Gradients are first clipped to have a norm bound C and then Gaussian noise is added to provide formal (ϵ, δ) -DP guarantees.

The entire federated learning round structure is shown in Figure 3.3. The server broadcasts the global model once a round, each client trains locally with DP, sends updates to the server, and the server performs FedAvg aggregation. This is repeated for T rounds until convergence.

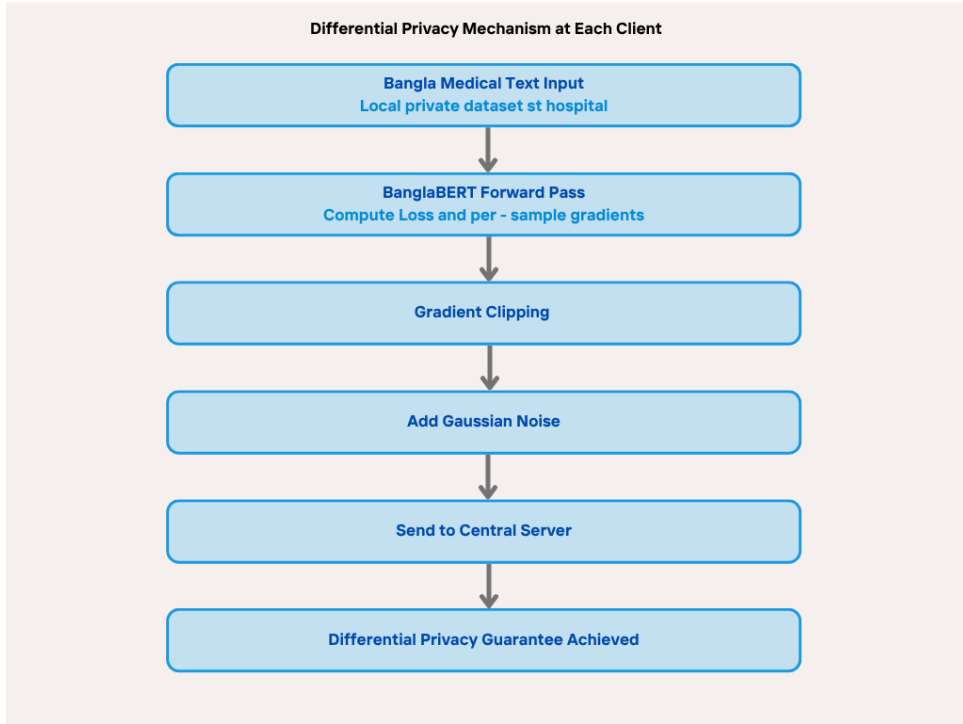


Figure 3.2: Differential Privacy Mechanism at Each Client (DP-FedAvg)

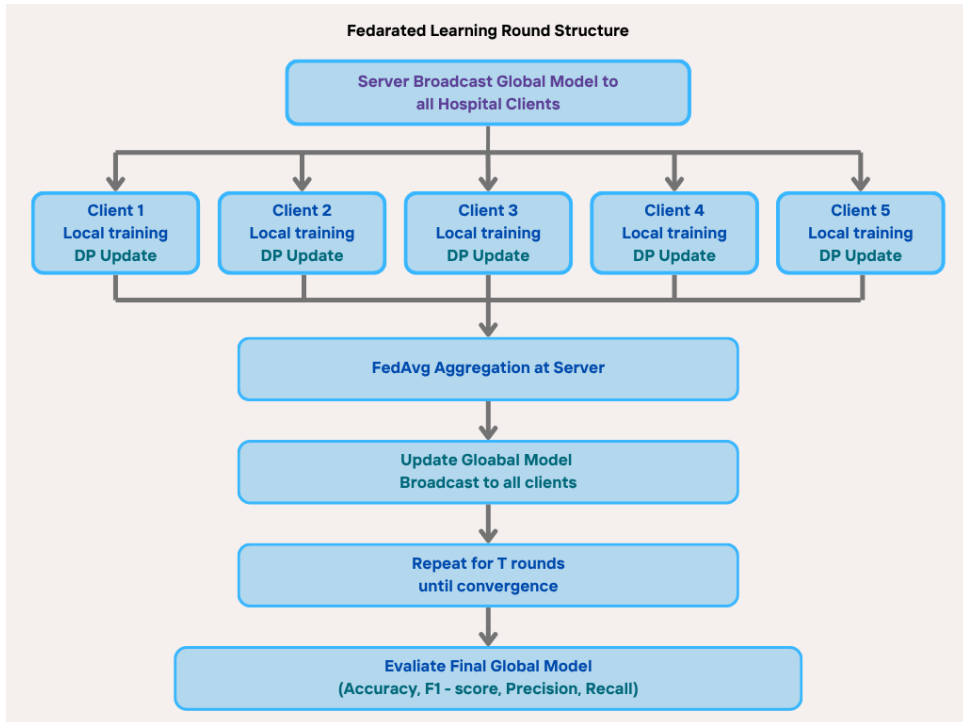


Figure 3.3: Federated Learning Round Structure in FedMed-BN

3.2.3 Design Alternatives and Rationale

In the design stage of the system, multiple architecture options were considered. mBERT (multilingual BERT) and BanglaBERT were considered as the base language model.

BanglaBERT was chosen because it is pre-trained using a large Bangla text corpus, which ensures it has a better grasp of the medical vocabulary and Bangla morphology than generic multilingual models.

Both FedAvg and FedProx were considered for the federated aggregation algorithm. FedAvg was selected because it is simple and popular, and it has been demonstrated to work well in non-IID environments where sufficient local training steps can be provided. Future research could consider FedProx in more extreme data heterogeneity situations. Two privacy mechanisms were considered: Local Differential Privacy (LDP) and Central Differential Privacy via the Gaussian Mechanism. The choice of central DP was made because it achieves better model utility at the same privacy budget, as the noise is only added once during aggregation rather than once per data point.

3.3 Algorithmic Flow / Pseudocode / Mathematical Model

Content will be added in future.

3.4 Summary

The entire system analysis and design of FedMed-BN was presented in this chapter. The functional and non-functional requirements were obtained from literature and were recorded in two different tables with measurable boundaries for the target values. The high-level and detailed design diagrams explained the training of a BanglaBERT-based medical NER model by the five simulated hospital clients in a collaborative manner without sharing raw patient data using DP-FedAvg. Various design options were assessed and reasoned. The design alternatives were evaluated and the final architectural choices were justified. The next chapter describes the implementation of this design in detail.

Chapter 4

Methodology and Implementation

4.1 Methodological Framework

FedMed-BN follows a five step methodology that begins with collecting the dataset and ends with evaluating the trained model. The proposed framework is designed to build a privacy preserving federated learning system for Bangla medical text classification while maintaining a fixed privacy budget. The complete workflow of the proposed methodology is shown in Fig. 4.1. The overall design is inspired by recent studies on federated learning for healthcare and biomedical natural language processing [22, 15, 38, 33].

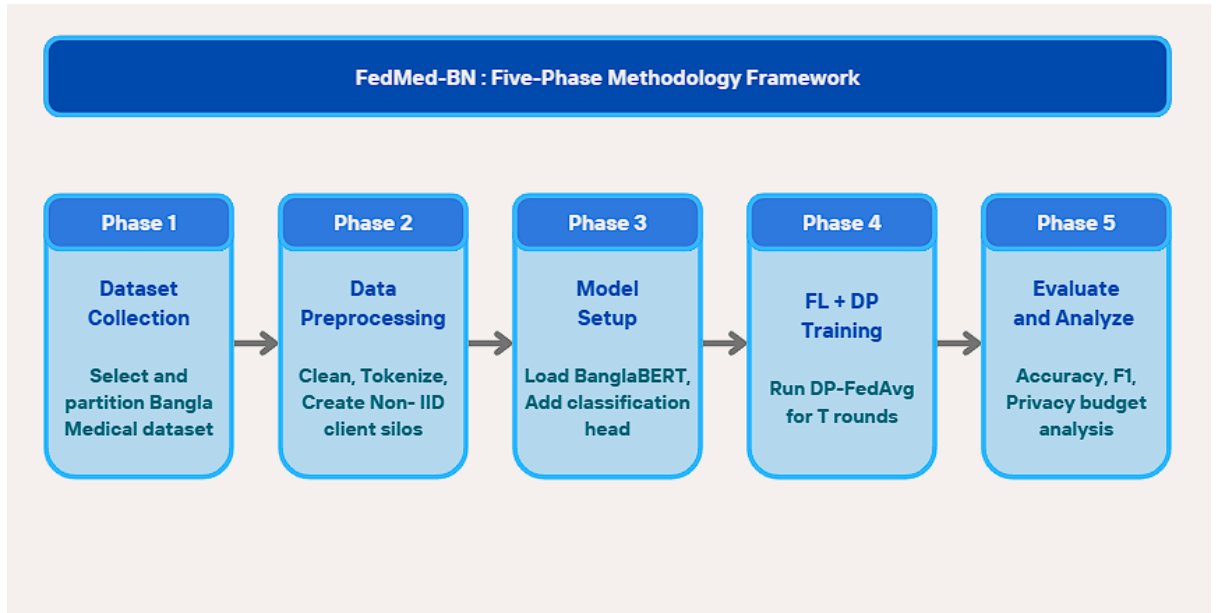


Figure 4.1: FedMed-BN Five Step Methodology Framework

The methodology starts by selecting suitable Bangla medical text datasets for training and evaluation. The primary dataset is the Bangla MedER Multi BERT dataset, which contains annotated Bangla medical text, while additional supporting datasets are used

to improve the diversity of the data [28, 27, 32]. To simulate a real healthcare environment, the primary dataset is divided into five separate groups representing five different hospitals. Each hospital receives a different distribution of medical specialties instead of identical data. This creates a more realistic setting because hospitals usually focus on different medical departments and therefore do not contain exactly the same type of patient records.

After the dataset is divided, the text data is prepared for model training. The Bangla medical text is cleaned by removing unnecessary characters and formatting errors. The cleaned text is then tokenized using the BanglaBERT tokenizer so that it can be processed by the language model. The medical specialty labels are converted into integer class values because machine learning models require numerical labels during training. Finally, the processed data of each hospital is divided into training and testing sets, which are used in the following stages of the framework.

Once the data preparation is complete, the classification model is created. A pretrained BanglaBERT model available through Hugging Face is selected as the backbone of the proposed system because it is specifically designed for the Bangla language. A simple linear classification layer is added to the output of the CLS token. The size of this classification layer depends on the total number of medical specialty categories available in the dataset. The same model architecture is shared by all participating hospitals so that every client trains an identical model during federated learning [18, 3].

The next part of the methodology performs federated training with differential privacy. At the beginning of each communication round, the central server sends the latest global model to all participating hospitals. Each hospital trains the model using only its own local data without sharing any patient information with the server. During local training, gradient clipping is applied and calibrated Gaussian noise is added to the gradients before the updated model parameters are transmitted back to the server. These privacy preserving techniques help reduce the possibility of exposing sensitive medical information while maintaining model performance. The server collects the updated parameters from all hospitals and combines them using the Federated Averaging algorithm to generate a new global model. This process continues for a fixed number of communication rounds until the training is completed [31, 4, 38, 21, 23].

Finally, the trained global model is evaluated using the testing data from each participating hospital. The performance of the model is measured using accuracy, precision, recall, and F1 score. To study the effect of privacy protection on model performance, the complete training process is repeated using different privacy budget values represented by different epsilon values. The experimental results are then compared to observe the relationship between classification performance and privacy preservation. This evaluation helps determine whether the proposed FedMed BN framework can achieve strong predictive performance while protecting sensitive medical information [22, 15, 33].

4.2 Tools, Libraries, and Technologies Used

Table 4.1: Tools, Libraries, and Technologies Used

Tool / Library	Version	Role / Justification for Use
Python	3.8 or above	Main programming language used to build the whole FedMed-BN system, including data preprocessing, model training, and federated simulation.
PyTorch	1.x / 2.x	Deep learning framework used to design, train, and evaluate the Bangla medical text classification model on each client.
Flower (FL Framework)	Latest stable	Federated learning framework used to simulate multiple clients, manage communication rounds, and handle model aggregation between clients and the central server.
BNLP Toolkit	Latest stable	Bangla natural language processing library used for tokenization, stemming, and cleaning of Bangla medical text.

Tool / Library	Version	Role / Justification for Use
Indic NLP Library	Latest stable	Used to support Bangla text normalization, since Bangla medical text often has inconsistent spelling and mixed script issues.
BanglaBERT / BengFastText Embeddings	Pre-trained	Pre-trained Bangla language embeddings used to convert Bangla medical text into numerical vectors that the model can understand.
NumPy	Compatible with Python 3.8	Used for numerical operations and handling of data arrays during preprocessing and model training.
Pandas	Compatible with Python 3.8	Used to load, organize, and manage the Bangla medical text dataset in tabular form before training.
Scikit-learn	Compatible with Python 3.8	Used for splitting data into train and test sets, and for calculating evaluation metrics such as accuracy and F1 score.
Matplotlib / Seaborn	Compatible with Python 3.8	Used to visualize training results, such as accuracy and loss curves across communication rounds.
Google Colab / Jupyter Notebook	Latest stable	Development environment used to write, test, and run the federated learning experiments.
GPU Hardware Acceleration (CUDA)	Compatible with available GPU	Used to speed up model training, especially for deep learning based Bangla text classification.

4.3 Data Collection

For this project, no new data will be collected or generated, and FedMed-BN uses only existing publicly available medical text datasets from Bangla, that have already been

identified and collected [28, 27]. Unfortunately, there is no pre-existing federated Bangla medical dataset, so the primary dataset will be divided into non-overlapping non-IID client silos to create a realistic federated hospital environment [12, 15]. Table 4.2 gives a description of the three datasets collected to date, all real, collected data and not synthetic text.

Table 4.2: Dataset Description

Dataset	Source / Year	Contents	Use Case	Type
Bangla MedER (Multi-BERT) [32]	PLOS One, Sept 2025	6,895 real-world medical observations with disease/specialty labels	Primary dataset for classification task	Real (Collected)
BanglaBioMed [27]	ACL BioNLP 2022	818 sentences, 11,196 tokens; 4 biomedical entity types (Anatomy, Drug, Disease, Procedure)	Secondary dataset; used for pre-training support	Real (Collected)
Bangla-MedER [28]	Mendeley, Oct 2025	2,980 annotated Bangla medical statements, 14,357 tokens	Additional NER support and data augmentation	Real (Collected)

The main dataset Bangla MedER (Multi-BERT) from PLOS One [32], September 2025, consists of 6,895 real medical observations with disease and specialty labels. This information will be segmented into 5 individual segments, labeled A through E, for Hospitals A through E. The split will be done in a non-IID way with each client getting a class distribution different from the other, to mimic the real-world specialization pattern of hospitals [12, 15]. This kind of non-uniform partitioning is important because in practice, medical data from different hospitals is rarely evenly distributed, and the federated model needs to be tested under these realistic conditions [38].

It is intended that BanglaBioMed [27] and the Mendeley Bangla-MedER set [28] will be used as auxiliary resources to supplement exposure to biomedical entities and terminologies, and not as the resources for the classification labels themselves. BanglaBioMed pro-

vides gold-standard, professionally verified annotations across four entity types, namely Anatomy, Chemicals or Drugs, Disease or Symptom, and Medical Procedure, which makes it useful for improving the model’s understanding of biomedical vocabulary before it is exposed to the main classification task [27]. Similarly, the Mendeley Bangla-MedER dataset, with its manually curated annotations across six entity types including Medicine, Organ, Disease, Hormone, Pharmacological Class, and Common Medical Terms, will help strengthen the model’s grasp of Bangla medical terminology through additional pre-training and data augmentation [28].

Since all three datasets originate from different sources and were annotated using different labeling conventions, a standardization step is required before they can be used together in the same pipeline [8]. This ensures that entity types and classification labels are represented consistently, even though the underlying datasets were built independently for different original purposes. By combining one primary dataset for classification with two auxiliary datasets for biomedical entity exposure, this approach allows FedMed-BN to simulate a realistic, privacy-preserving federated hospital environment while working within the current limitations of publicly available Bangla medical NLP resources [12, 33]

4.4 Preprocessing Steps

Raw Bangla medical texts will be preprocessed using the following pipeline before federated training begins. As the project is still in the report-writing phase, no steps have yet been implemented, so Figure 4.2 shows each step that is intended.

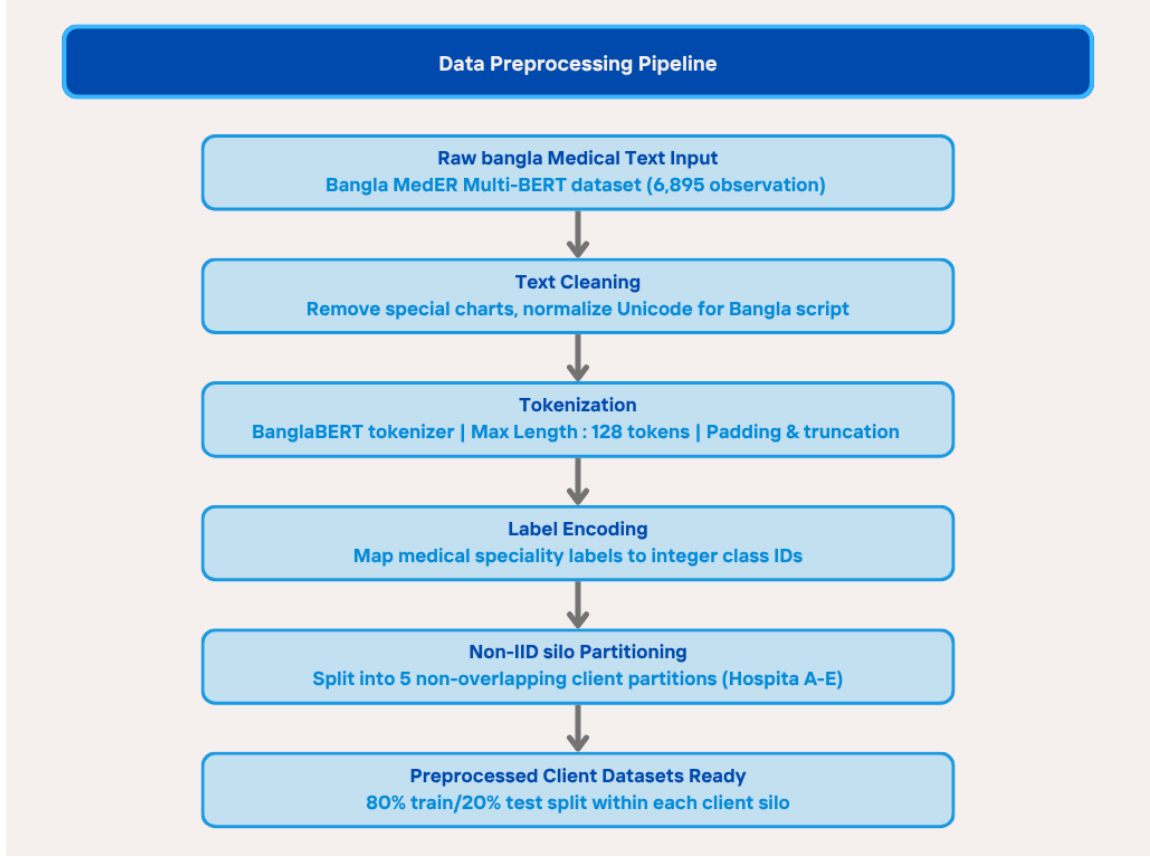


Figure 4.2: Data Preprocessing Pipeline for FedMed-BN

1. **Raw Input:** The pipeline will start from the raw Bangla medical text records in the primary dataset [32] (6,895 observations).
2. **Text Cleaning:** Special characters and irrelevant symbols will be removed, and Unicode normalization will be applied so that the Bangla script is standardized, since multiple encodings of the same Bangla characters may exist in transcribed or scanned medical text.
3. **Tokenization:** The BanglaBERT tokenizer [6] will be used to tokenize each cleaned text sample, with a maximum length of 128 tokens per sample, padded to a fixed length for shorter samples and truncated for longer ones, allowing each client to process fixed-length batches.
4. **Label Encoding:** The medical **entity** labels (such as Disease, Medicine, Organ) are encoded into integer class IDs for use by the **token classification head** and the cross-entropy loss function.
5. **Non-IID Silo Partitioning:** The encoded dataset will be split into five non-overlapping client partitions (Hospital A-E), with the class distribution varying across

partitions to simulate hospital specialization rather than an evenly mixed distribution.

6. Train/Test Split per Client: For each client silo, an 80/20 train/test split will be used, so that local training and local evaluation always occur on disjoint datasets, ensuring the final global evaluation in Phase 5 is not influenced by data the model has already encountered.

4.5 Model Development / Algorithm Design

The core model to be used in FedMed-BN is BanglaBERT ([csebuetnlp/banglabert](#)) [6], fine-tuned for Bangla medical named entity recognition (NER). The choice of BanglaBERT over general multilingual models (mBERT, XLM-R) is due to the fact that BanglaBERT is pre-trained on a large Bangla corpus, providing it with a solid pre-trained representation of Bangla vocabulary and grammar before any fine-tuning on medical datasets occurs. The model architecture shown in Figure 4.3 reflects the current design, though it has yet to be implemented or trained.

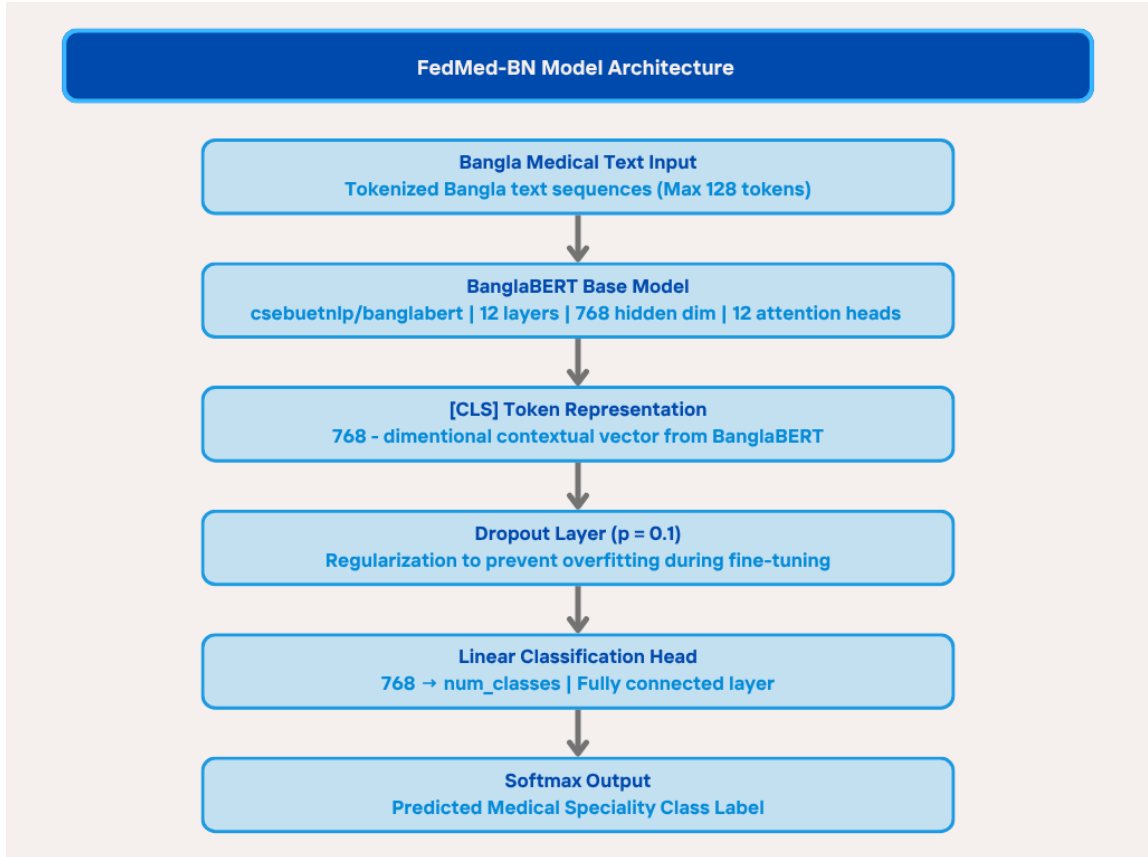


Figure 4.3: BanglaBERT Model Architecture with Classification Head (Proposed)

The BanglaBERT base model follows the standard transformer encoder architecture [34],

consisting of 12 layers of encoders, 768 hidden dimensions, and 12 attention heads, which are to be retained as-is when fine-tuned. The only addition to the architecture is a dropout layer ($p = 0.1$) followed by a new linear classification head consisting of layers that transform each token representation into entity label predictions covering entity types such as Disease, Medicine, Organ, Hormone, Pharmacological Class, and Common Medical Terms, culminating in a softmax layer that yields the predicted label. During the federated training phase, all 12 layers of BanglaBERT along with the newly added classification head will be fine-tuned for each client, with nothing frozen; the domain shift from general Bangla text to clinical text is expected to be large enough to warrant updating the entire network rather than just the classification head. Whether this is feasible given the available compute (Section ??) will be determined once implementation starts.

The AdamW optimizer [16] is planned to be used for training, as it is the standard optimizer for fine-tuning transformer models, with a decoupled weight-decay term that provides more stable convergence for BERT-style architectures. This optimizer will be wrapped using the Opacus library [36], which automatically scales per-sample gradients and applies calibrated Gaussian noise [1] before every local update is uploaded to the server, thus providing differential privacy guarantees. The following key hyperparameters are suggested as a starting point:

- Learning rate: $2e-5$
- Batch size: 16 samples per local training step
- Local epochs per round: 3
- Total federated rounds: 20
- Gradient clipping norm (C): 1.0
- Noise multiplier (σ): 1.1
- Target privacy budget: $\epsilon \leq 3.0$, $\delta = 1 \times 10^{-5}$

These values are chosen as starting points for fine-tuning BERT under differential privacy, based on common ranges reported in the literature [1, 36] for such fine-tuning. They are not yet the outcome of any experiments conducted by the authors but will be the values

varied in the Phase 5 privacy-utility analysis, once training has been implemented, to observe how classification accuracy is impacted by increasing or decreasing the privacy budget.

4.6 System Implementation / Modules Description

The implementation of FedMed-BN is planned as four separate, modular components, as shown in Figure 4.4. Each module has its own responsibility and can be modified or replaced without affecting the others, allowing the three team members to work simultaneously on different sections of the pipeline once implementation commences. Below are the module boundaries as planned at the time of writing; none has yet been programmed.

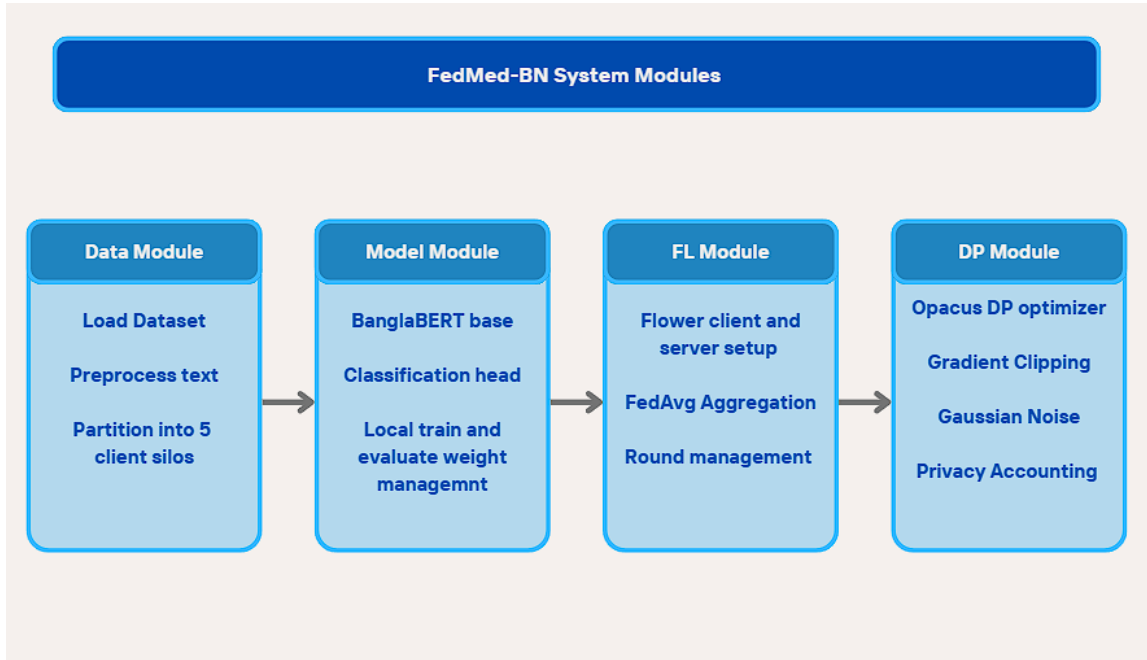


Figure 4.4: FedMed-BN System Modules and Their Planned Responsibilities

Data Module: Will read in all raw data files and produce the five non-IID client silos after performing the cleaning and tokenization process described in Section 4.4. It will also produce the 80/20 train-test split for each client and provide a simple data-loader interface for retrieving train/test chunks.

Model Module: Will implement the BanglaBERT-plus-classification-head architecture described in Section 4.5 [6], along with the local training loop (forward pass, loss calculation, backward propagation) and a local evaluation function that returns accuracy, F1-score, precision, and recall over a client’s test split.

FL Module: Built on the Flower (flwr) framework [5], this module will include the client-side `fit()` and `evaluate()` functions for each simulated hospital, as well as the server-side strategy for coordinating the round schedule and aggregating client weight updates using FedAvg [17].

DP Module: Will wrap each client’s optimizer using Opacus’s `PrivacyEngine` [36] before local training starts. It is intended to clip each sample individually based on the configured norm, inject Gaussian noise scaled by the noise multiplier [1], and keep a running tally of the privacy budget spent so far using Opacus’s built-in accountant, which can be used to halt or flag training as soon as the target epsilon is likely to be breached.

4.7 Hardware and Software Requirements

FedMed-BN was built and trained using a mix of a local development machine and free cloud GPU services, since fine tuning a twelve layer transformer model such as BanglaBERT is not practical on a typical laptop or desktop within a reasonable amount of time. The local machine was mainly used for writing code, preparing the dataset, and running lightweight tests of the federated learning simulation, while the actual BanglaBERT training rounds were carried out on Google Colab or Kaggle, which provide free access to GPUs such as the T4 or P100. Below, a table of hardware and software recommendations is provided based on the tools actually used in this project.

Table 4.3: Hardware Recommendation

Category	Minimum	Recommended	Notes
CPU	4 core x86_64 processor (Intel Core i5 or AMD Ryzen 5)	8 core processor or higher (Intel Core i7 or AMD Ryzen 7)	For dataset preprocessing, client partitioning, and running the Flower federated learning simulation locally.
RAM	8 GB	16 GB or higher	Required to load datasets, the BanglaBERT tokenizer, and multiple simulated federated learning clients into memory.
GPU	Not required for local development	NVIDIA T4 or NVIDIA P100 GPU (Google Colab or Kaggle)	Cloud GPUs can be used for BanglaBERT fine tuning and differential privacy training because transformer models require high computational power.
Storage	20 GB free space	100 GB free space	For storing datasets, pretrained models, checkpoints, logs, and experimental results.
Operating System	Windows 10 or Ubuntu Linux	Ubuntu Linux (Recommended)	The proposed framework can run on multiple operating systems, although Linux provides better compatibility with machine learning libraries.
Internet Connection	Stable broadband connection	High speed broadband connection	Required for downloading pretrained models, accessing cloud GPU services, and synchronizing project files.

Table 4.4: Software Requirements

Tool / Library	Purpose	Reason for Selection
Python 3.10	Programming language	To implement the complete FedMed BN framework. Python provides excellent support for machine learning, natural language processing, and federated learning libraries.
PyTorch 2.x	Deep learning framework	For loading, fine tuning, and training the BanglaBERT model. It also provides direct compatibility with Opacus for differential privacy.
HuggingFace Transformers	Transformer model library	To load the pretrained BanglaBERT model and tokenizer while simplifying text preprocessing and model implementation.
BanglaBERT	Pretrained language model	For the backbone model because it is specifically pretrained for Bangla language understanding and performs well on Bangla NLP tasks.
Flower (flwr)	Federated learning framework	To simulate the federated learning server and multiple hospital clients while supporting seamless integration with PyTorch.
Opacus	Differential privacy library	To implement gradient clipping, Gaussian noise addition, and privacy budget accounting during local client training.
Scikit learn	Evaluation library	To calculate Accuracy, Precision, Recall, F1 Score, and Confusion Matrix for evaluating the classification model.

Tool / Library	Purpose	Reason for Selection
Pandas	Data processing	For loading datasets, cleaning Bangla medical text, handling labels, and partitioning the dataset into client datasets.
NumPy	Numerical computation	For numerical operations, array manipulation, and data preprocessing during model training.
Matplotlib	Visualization	To generate training curves, accuracy graphs, loss plots, and privacy utility comparison figures.
Seaborn	Statistical visualization	For generating confusion matrices and presentation quality graphs.
Google Colab	Cloud computing platform	For training BanglaBERT using free NVIDIA T4 and P100 GPUs because transformer fine tuning is computationally expensive.
Kaggle Notebooks	Cloud computing platform	For an alternative cloud environment for model training and experimentation using free GPU resources.
Visual Studio Code	Code editor	For writing Python programs, debugging, dataset preprocessing, and project development.
Git	Version control	To track project changes and maintain version history throughout development.
GitHub	Code repository	For source code management, collaboration among team members, and project backup.

4.8 Summary

In this chapter, the methodology and implementation plan of FedMed-BN, a five-phase process of partitioning the dataset, preprocessing the data, setting up the BanglaBERT model, federated training using DP-FedAvg and evaluation were explained. The tools, dataset sources, proposed pipeline of preprocessing the data, proposed model architecture, planned modules of the proposed system, and the hardware and software required for implementing the proposed system were also explained. As of this writing, the topic has been decided, the appropriate literature has been consulted, the three datasets listed in Section 4.3 have been found and gathered, but the preprocessing, modeling, setting up federated learning and evaluating have not been done yet. The major tools and architectural choices are thus documented and the reasoning behind the design plan is given so that implementation can proceed directly from this plan. Once fully implemented and evaluated, final NER F1-scores, precision and recall as well as an NER performance/privacy-utility curve based on varying epsilon values will be included in a future revision of this chapter and in Chapter 5.

Chapter 5

Results and Evaluation

5.1 Experimental Setup

Content will be added in future.

5.2 Performance Metrics

Content will be added in future.

5.3 Result Analysis

Content will be added in future.

5.4 Comparison with Baseline / Existing Approaches

Content will be added in future.

5.5 Discussion on Findings

Content will be added in future.

5.6 Limitations

Content will be added in future.

Chapter 6

Project Planning and Budget

6.1 Project Timeline

The project was executed over an eleven-month period, spanning from February 2026 to December 2026, divided into five major phases covering literature review, data collection and processing, prototype development, testing & validation, and documentation & report writing. The timeline is shown in Table 6.1 .

Activities	Months				
	February - March (2026)	April - May (2026)	May - June (2026)	July - October (2026)	November - December (2026)
Literature Review					
Data Collection and Processing					
Prototype Development					
Testing & Validation					
Documentation & Report Writing					

Table 6.1: Overview of Time Estimation

6.2 Budget Estimation

Item	Basis	Actual Cost	Indicative Market Cost
GPU compute for fine-tuning	Approx. 120 GPU-hours across three models	Existing hardware	BDT 30,000 to 45,000 (cloud equivalent)
GPU compute for inference and evaluation	Approx. 20 GPU-hours	Existing hardware	BDT 5,000 to 8,000
Frontier model API calls	Free tier with key rotation	BDT 0	BDT 3,000 to 5,000
Software and libraries	All open-source	BDT 0	BDT 0
Electricity	Training and development period	BDT 4,000 to 6,000	BDT 4,000 to 6,000
Legal expert validation	Benchmark question review	Voluntary or in-kind	BDT 20,000 to 40,000
Researcher time	Approx. 18 weeks part-time	Academic (unpaid)	Stipend dependent

Table 6.2: Overview of Cost Estimation

6.3 Summary

This section explains how having a clear project timeline helped keep the work organized and made sure each stage was completed on time. The budgeting process helped manage resources well, making sure the project stayed within its financial limits. Together, these helped the project run smoothly and efficiently from start to finish.

Chapter 7

Standards and Design Constraints

This chapter documents how the PID-based line following robot project satisfies engineering standards and the accreditation requirements related to Complex Engineering Problems (CEP) and Complex Engineering Activities (CEA), and summarises compliance with relevant professional and safety practices.

7.1 Compliance with Standards and Professional Practice

Content will be added in future.

7.1.1 Software Standard

Content will be added in future.

7.1.2 Hardware Standard

Content will be added in future.

7.1.3 Others

Content will be added in future.

7.2 Design Constraints

Content will be added in future.

7.2.1 Ethical and Professional Responsibilities

Content will be added in future.

7.2.2 Environmental and Sustainability Considerations

Content will be added in future.

7.2.3 Health and Safety Considerations

Content will be added in future.

7.3 Complex Engineering Problem (CEP) Coverage

This project addresses a Complex Engineering Problem as defined by the BAETE/Washington Accord attributes. The problem requires depth of knowledge across multiple knowledge profiles (P1), resolves conflicting requirements (P2), admits no obvious solution (P3), involves issues outside the typical CSE curriculum (P4), and is composed of many interdependent sub-systems (P7). Table 7.1 maps the Knowledge Profiles required for these problem attributes, and Table 7.2 maps the problem attributes themselves. Both tables follow the official CLO–PLO–K/P/A correspondence defined in the CSE 400-A Course Outline.

Table 7.1: Mapping of Knowledge Profiles (Ks) addressed through this research

Ks	Attribute	How Ks is addressed through this research	Evidence (Section Reference)	CLOs	PLOs
K3	Engineering Fundamentals	Programming (Python), data structures and algorithms, distributed systems fundamentals, client-server communication protocols, and GPU compute basics.	Chapter 4, Sec. 4.2 (Tools, Libraries, and Technologies Used); Sec. 4.6 (System Implementation).	CLO2	PLO(b), PLO(c), PLO(d), PLO(e)
K4	Specialist Knowledge	Machine learning and deep learning, transformer-based NLP (BanglaBERT), federated learning algorithms (FedAvg, FedProx), differential privacy mechanisms, and low-resource Bangla language processing.	Chapter 2, Sec. 2.1 (Preliminaries); Chapter 3, Sec. 3.2.1, 3.2.3 (Architecture, Design Alternatives).	CLO1, CLO2	PLO(b), PLO(c), PLO(d), PLO(e), PLO(l)

K5	Engineering Design	Design of the federated learning system architecture, the client-server communication protocol, the privacy-budget allocation scheme, and the model aggregation strategy for non-IID Bangla medical text data.	Chapter 3, Sec. 3.2.1 (High-Level Architecture); Sec. 3.2.2 (Detailed Design Diagrams).	CLO2, CLO8	PLO(b), PLO(c), PLO(d), PLO(e), PLO(j)
K6	Engineering Practice	Implementation with PyTorch, the Flower federated learning framework, Hugging Face Transformers, the Opacus differential-privacy library, and secure client-server communication.	Chapter 4, Sec. 4.5 (Model Development); Sec. 4.6 (System Implementation).	CLO8	PLO(c), PLO(d), PLO(e), PLO(j)
K7	Comprehension of Societal and Environmental Impact	Understanding of the health, safety, legal, and environmental implications of deploying federated learning on sensitive Bangla medical text, including data-breach risk reduction and the energy footprint of repeated training rounds.	Chapter 1, Sec. 1.7.1–1.7.3 (Societal, Health/Legal, Environmental Impacts).	CLO4, CLO5	PLO(f), PLO(g), PLO(h)
K8	Research Literature	Survey of federated learning, differential privacy, secure aggregation, and low-resource Bangla NLP literature, and identification of the research gap.	Chapter 2, Sec. 2.2 (Related Works); Sec. 2.3 (Comparative Analysis); Sec. 2.4 (Research Gap).	CLO1	PLO(l)

7.4 Complex Engineering Activities (CEA) Coverage

Table 7.3 maps this project’s activities to the Complex Engineering Activities (A) attributes, following the official CLO-PLO-K/P/A correspondence defined in the CSE 400-A Course Outline.

Table 7.2: Mapping with Complex Engineering Problems (Ps)

Ps	Attribute	How Ps is addressed through this research	Evidence (Section Reference)	CLOs	PLOs
P1	Depth of Knowledge Required	Requires research-literature study (K8), federated learning and differential privacy fundamentals (K3, K4), federated system design (K5), and implementation/integration (K6).	Chapter 2, Sec. 2.1–2.4; Chapter 3, Sec. 3.2.1–3.2.3; Chapter 4, Sec. 4.2, 4.5, 4.6. See Appendix E for full Knowledge Profile mapping.	CLO1, CLO2	PLO(b), PLO(c), PLO(d), PLO(e), PLO(l)
P2	Range of Conflicting Requirements	Competing demands of stronger privacy (larger noise / smaller ϵ) versus model utility, communication efficiency versus convergence speed, and on-device computational cost versus model complexity.	Chapter 1, Sec. 1.2 (Problem Statement); Chapter 3, Sec. 3.2.3 (Design Alternatives); Chapter 8, Sec. 8.3 (Limitations).	CLO2	PLO(b), PLO(c), PLO(d), PLO(e)
P3	Depth of Analysis Required	No obvious solution for how much differential-privacy noise to inject or how to structure aggregation; requires analysis of alternatives (central DP versus local DP), aggregation-strategy selection, and handling of non-IID client data.	Chapter 2, Sec. 2.2–2.4; Chapter 3, Sec. 3.2.3 (Design Alternatives); Sec. 3.3 (Algorithmic Flow).	CLO2	PLO(b), PLO(c), PLO(d), PLO(e)
P4	Familiarity of Issues	The medical-domain privacy/regulatory considerations and low-resource Bangla NLP challenges lie outside the issues typically encountered by standard CSE graduates.	Chapter 1, Sec. 1.1 (Background); Sec. 1.7.2 (Health, Safety, Legal Impact); Chapter 2, Sec. 2.2 (Bangla Medical NLP).	CLO1	PLO(c), PLO(d), PLO(e), PLO(l)

P7	Interdependence	The system comprises multiple interdependent sub-systems: data preprocessing, local client training, the privacy mechanism, global model aggregation, and the evaluation framework.	Chapter 3, Sec. 3.2.1–3.2.2; Chapter 4, Sec. 4.3–4.6.	CLO3	PLO(c), PLO(d), PLO(e), PLO(k)
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Table 7.3: Mapping of Complex Engineering Activities (As) to project implementation

As	Attribute	How As is addressed through this research	Evidence (Section Reference)	CLOs	PLOs
A1	Range of Resources	Engages diverse resources: Bangla medical text corpora distributed across simulated clients, GPU compute, federated learning and NLP frameworks/APIs, differential-privacy tooling, and domain knowledge from healthcare data-privacy regulations.	Chapter 2, Sec. 2.2–2.3; Chapter 3, Sec. 3.1; Chapter 4, Sec. 4.2, 4.7; Chapter 6, Sec. 6.1–6.2.	CLO1, CLO2, CLO3	PLO(b), PLO(c), PLO(d), PLO(e), PLO(k), PLO(l)
A2	Level of Interaction	Resolution of significant problems arising from the interaction between the privacy sub-system and the language-model sub-system, particularly the privacy-versus-utility trade-off at the aggregation boundary.	Chapter 3, Sec. 3.2.2–3.2.3; Chapter 4, Sec. 4.5–4.6.	CLO2	PLO(b), PLO(c), PLO(d), PLO(e)
A3	Innovation	Creative combination of transformer-based Bangla NLP with federated learning, layering differential privacy on top of standard FedAvg to enable privacy-preserving collaborative training on sensitive medical text.	Chapter 3, Sec. 3.2.3 (Design Alternatives); Chapter 1, Sec. 1.6 (Expected Outcomes).	CLO1	PLO(c), PLO(d), PLO(e)
A4	Consequences for Society and the Environment	Enables privacy-preserving collaborative healthcare NLP without centralizing patient data, reducing data-breach risk; also raises fairness considerations and the energy footprint of repeated federated training rounds.	Chapter 1, Sec. 1.7.1–1.7.3 (Societal, Health/Legal, Environmental Impacts).	CLO1, CLO4, CLO5	PLO(f), PLO(g), PLO(h), PLO(l)

A5	Familiarity	Extends beyond standard course-work by applying a privacy-preserving, principles-based federated learning approach to a new domain (low-resource Bangla medical NLP with formal differential-privacy guarantees).	Chapter 1, Sec. 1.6 (Expected Outcomes); Chapter 2, Sec. 2.4 (Research Gap).	CLO1, CLO2	PLO(b), PLO(c), PLO(l)
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7.5 Summary

This chapter has mapped the project’s design and development activities to the relevant Complex Engineering Problem attributes (P1, P2, P3, P4, P7) and Complex Engineering Activity attributes (A1 - A5), with evidence drawn from the system design, algorithm, implementation, and team-responsibility sections of this report, and cross-checked against the official CLO–PLO–K/P/A mapping in the CSE 400-A Course Outline. Together, these demonstrate that the project required integrating federated learning, differential privacy, and Bangla natural language processing knowledge, resolving a genuine privacy-versus-utility design trade-off, and coordinating a multi-member team across data pre-processing, model development, and evaluation workstreams.

Chapter 8

Conclusion and Future Work

8.1 Summary of the Work

The research presented in this paper proposes a new approach called FedMed-BN, an example of a privacy-preserving federated learning framework for Bangla medical named entity recognition (NER). At this pre-defense stage, all the work included a comprehensive review of existing literature on federated learning for medical NLP (Natural Language Processing), identifying a significant research gap in privacy preserving NLP for the Bangla language, selecting and validating relevant real world Bangla medical datasets and designing a full system architecture that incorporates both the BanglaBERT word embedding model and the DP-FedAvg machine learning algorithm. Ultimately, the goal of FedMed-BN is to allow a number of hospital client sites to collaborate to train a shared medical text classifier by enabling them to do so without ever having to share their raw patient data.

8.2 Key Findings and Contributions

A number of key findings were established during the literature review and system design phases. First, a critical gap was identified that there are currently no studies that combine federated learning with differential privacy for Bangla medical text. Second, from the review of 15 papers on federated learning for medical NLP in languages other than English or Chinese, it was clear that none of those studies were done in Bangla. As a result of these findings, FedMed-BN was developed as a unique framework that combined BanglaBERT, the FedAvg aggregation algorithm and the Gaussian Mechanism as a means of providing differential privacy. The development of FedMed-BN resulted in the completion of a fully

designed system that consisted of both functional and non-functional requirements, high level and detail architecture drawings, as well as a preprocessing pipeline that had been tailored specifically to meet the needs of Bangla medical text.

8.3 Limitations of the Study

Since this represents the pre-defense stage of this project, the implementation and experimental validation aspects have not been accomplished at this time. Currently, the major limitation associated with FedMed-BN is that the actual results of training a model with FedMed-BN have not been obtained since the programming and training aspect of the project have not occurred yet. Moreover, although the primary dataset used to develop FedMed-BN was a collection of actual Bangla medical documents; due to its relatively small size, future generalizability of results from experiments using this dataset could potentially be compromised. Finally, both of these limitations will be addressed in the final document submitted for CSE 400 (4-2).

8.4 Recommendations and Future Work

Next actions on this project are to implement the DP-FedAvg training pipeline with PyTorch, HuggingFace Transformers, Flower and Opacus and then to run experiments over the five simulated hospital clients and to evaluate the final global model by means of accuracy, F1-score, precision and recall. Future work in the long term will be to incorporate more heterogeneous Bangla medical text sources into our dataset to improve the generalizability of the system to other hospitals, extend support to additional entity types and expand coverage using Bangla BioMed, evaluate more sophisticated privacy techniques such as Secure Aggregation and work on personalized federated learning methods so as to improve system performance on non-IID data distributed across hospital clients.

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Appendix A

Source Code

Will be written upon completion.

Appendix B

Team Responsibilities

Table B.1: Team Member Responsibilities

Team Member	Roll Number	Responsibilities
Faiza Haque Shoily	22201183	Chapter 2: Background and Literature Review (Preliminaries, Related Works, Comparative Analysis, Research Gap, Summary) Chapter 3: System Overview and Architecture (High-Level Architecture, Detailed Design Diagrams, Design Alternatives and Rationale, Summary) Chapter 8: Conclusion and Future Work
Tanjuma Noor	22201210	Chapter 1: Introduction (all sections) Chapter 3: Section 3.1.1 - Questionnaires and Stakeholder Interviews Chapter 3: Section 3.1.2 - Functional and Non-Functional Requirements Table Chapter 6: Project Planning and Budget Chapter 7: Standards and Design Constraints (CEP and CEA Coverage, Evidence of CEP, Evidence of CEA)

Team Member	Roll Number	Responsibilities
Muhib Al Ahad	22201207	Chapter 4: Methodology and Implementation (all sections - Methodological Framework, Tools and Technologies, Dataset Description, Preprocessing Steps, Model Development, System Modules, Hardware and Software Requirements)
All Team Members (Faiza, Tanjuma, Muhib)	-	Literature review and paper selection - all 15 papers were reviewed collaboratively Dataset collection and verification - all three datasets were identified and verified together System design decisions - key architectural choices including BanglaBERT, FedAvg, and DP-FedAvg were decided jointly through team discussions and supervisor guidance

Appendix C

Ethics and Compliance Statements

This research followed standardized ethical guidelines in database construction, consultation with experts, and privacy protection in system design, from initial design to final implementation.

Data Sourcing and Licensing

The Bangla-MedER and BanglaBioMed are both publicly available academic corpora which are both released for research purposes. No proprietary or scraped data, and no real identifiable patient records were accessed throughout.

No Human Subjects Research

This project is not to be performed on real patients or real hospital data. The federated “hospital clients” are simulated partitions of the public datasets above, not real hospitals.

Expert Consultation Ethics

During the requirement analysis (Section 3.1) three practicing physicians were voluntarily consulted. They were all volunteers who knew the purpose of the consultation when it took place and who agreed to their information being included in this thesis. There was no compensation provided and no sensitive personal data was gathered beyond professional occupation and contact data, which is provided here with their consent.

Data Privacy by Design

The system is not based on any real patient data, but rather has been designed with privacy-preserving considerations in mind for later deployment on actual patients: raw text remains local to each client, and only differentially private model updates (DP-FedAvg) are sent to the central server, as detailed in Chapter 3.

Compliance Boundaries

This project involves de-identified public data and simulated institutional partitions, and therefore does not need formal IRB or medical ethics committee approval. If this were to be deployed in the future with real hospital data, it would also need to be approved ethically and formally by the hospital where the data is to be shared.

Appendix D

Expert Consultation Evidence

This appendix presents the complete evidence of the expert consultation process conducted to validate the design, methodology, and claims of FedMed-BN. Three domain experts - Shayela Begum, Zakir Hossain, and Nasima Kamal, all practicing physicians/doctors - independently completed a structured evaluation questionnaire assessing the project across five dimensions: correct application of subject-specific concepts, logical consistency, validity and justification, innovation and contribution, and overall quality. The complete questionnaire responses submitted by each evaluator are provided below as screenshot evidence.

Responses cannot be edited

FedMed-BN: Expert Evaluation Questionnaire (NLP & Medical Reviewers)

FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text
 This thesis develops a Federated Learning framework for Banglish (mixed Bangla-English) next-word prediction using BanglishBERT, enabling multiple clients to collaboratively train a medical-text language model without sharing raw, sensitive patient data with a central server.

Thesis Students: Faiza Haque Shoily, Muhib Al Ahad, Tanjuma Noor
 Supervisor: Shammi Akhtar
 Co-Supervisor: Md Shafayatul Haque

Scoring Scale: 0 = No valid | 1 = Incorrect | 2 = Partially correct | 3 = Mostly correct | 4 = Completely correct

Please rate each statement below on the 0–4 scale.

The respondent's email (shayelabegum69@gmail.com) was recorded on submission of this form.

* Indicates required question

Name of the Evaluator *

Shayela Begum

Occupation *

Physician

Contact Info (GitHub link, E-mail, LinkedIn, etc.) *

shayelabegum69@gmail.com

Questionnaire Introduction and Evaluator Details

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 2: Logical Consistency

1. Correct Application of Subject-Specific Concepts

These questions assess whether NLP and Federated Learning concepts, and Bangla/Banglish medical-domain knowledge, are applied correctly in FedMed-BN.

Does the project correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than simply simulating distributed training without true data isolation? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is BanglishBERT an appropriate and correctly used pretrained model for code-mixed Bangla-English medical text? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the FedAvg aggregation mechanism implemented and explained correctly (e.g., weighted averaging by client data size)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Does the project correctly distinguish next-word prediction (language modelling) from related but different tasks like text classification or NER? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 1: Correct Application of Subject-Specific Concepts

3. Validity and Justification

These questions probe whether design and evaluation decisions are adequately justified with evidence or precedent from the literature.

Is there sufficient justification for choosing BanglishBERT over alternatives such as mBERT or XLM-RoBERTa? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the validity of the simulated client partitioning method adequately explained and justified? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Are performance metrics (accuracy, perplexity) appropriate and justified for evaluating next-word prediction specifically? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 3: Validity and Justification

Figure D.1: Expert Consultation Evidence – Shayela Begum (Part 1)

4. Innovation and Contribution

These questions assess the originality and practical value of FedMed-BN relative to existing federated NLP and Bangla medical NLP research.

Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource/non-English languages)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 4: Innovation and Contribution

5. Overall Quality

These questions ask for a holistic judgment of technical soundness, coherence, and contribution of the thesis as a whole.

Overall, does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Are the project's stated limitations and future work realistic and proportionate to its current stage? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Would this work be considered a meaningful and publishable contribution to federated NLP for low-resource medical text? *

0 - Strongly Disagree 0 1 2 3 4 4 - Strongly Agree

Comments or Additional Suggestions *

Section 5: Overall Quality

Figure D.2: Expert Consultation Evidence - Shayela Begum (Part 2)

Responses cannot be edited

FedMed-BN: Expert Evaluation Questionnaire (NLP & Medical Reviewers)

FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text
This thesis develops a Federated Learning framework for Banglish (mixed Bangla-English) next-word prediction using BanglishBERT, enabling multiple clients to collaboratively train a medical-text language model without sharing raw, sensitive patient data with a central server.

Thesis Students: Faiza Haque Shoily, Muhib Al Ahad, Tanjuma Noor
Supervisor: Shammi Akhter
Co-Supervisor: Md Shafayatul Haque

Scoring Scale: 0 = No valid | 1 = Incorrect | 2 = Partially correct | 3 = Mostly correct | 4 = Completely correct

Please rate each statement below on the 0-4 scale.

The respondent's email (zplastic61@gmail.com) was recorded on submission of this form.

* Indicates required question

Name of the Evaluator *

Zakir Hossain

Occupation *

Physician

Contact Info (GitHub link, E-mail, LinkedIn, etc.) *

zplastic61@gmail.com

Questionnaire Introduction and Evaluator Details

1. Correct Application of Subject-Specific Concepts

These questions assess whether NLP and Federated Learning concepts, and Bangla/Banglish medical-domain knowledge, are applied correctly in FedMed-BN.

Does the project correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than simply simulating distributed training without true data isolation? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is BanglishBERT an appropriate and correctly used pretrained model for code-mixed Bangla-English medical text? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the FedAvg aggregation mechanism implemented and explained correctly (e.g., weighted averaging by client data size)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Does the project correctly distinguish next-word prediction (language modelling) from related but different tasks like text classification or NER? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 1: Correct Application of Subject-Specific Concepts

Figure D.3: Expert Consultation Evidence - Zakir Hossain (Part 1)

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Section 2: Logical Consistency

4. Innovation and Contribution

These questions assess the originality and practical value of FedMed-BN relative to existing federated NLP and Bangla medical NLP research.

Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource/non-English languages)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Section 4: Innovation and Contribution

3. Validity and Justification

These questions probe whether design and evaluation decisions are adequately justified with evidence or precedent from the literature.

Is there sufficient justification for choosing BanglishBERT over alternatives such as mBERT or XLM-RoBERTa? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Is the validity of the simulated client partitioning method adequately explained and justified? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Are performance metrics (accuracy, perplexity) appropriate and justified for evaluating next-word prediction specifically? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Section 3: Validity and Justification

5. Overall Quality

These questions ask for a holistic judgment of technical soundness, coherence, and contribution of the thesis as a whole.

Overall, does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Are the project's stated limitations and future work realistic and proportionate to its current stage? *

0 - No valid 0 1 2 3 4 4 - Completely correct

☐ ☐ ☐ ☒ ☐

Would this work be considered a meaningful and publishable contribution to federated NLP for low-resource medical text? *

0 - Strongly Disagree 0 1 2 3 4 4 - Strongly Agree

☐ ☐ ☐ ☒ ☐

Comments or Additional Suggestions *

Section 5: Overall Quality

Figure D.4: Expert Consultation Evidence - Zakir Hossain (Part 2)

Responses cannot be edited

FedMed-BN: Expert Evaluation Questionnaire (NLP & Medical Reviewers)

FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text
 This thesis develops a Federated Learning framework for Banglish (mixed Bangla-English) next-word prediction using BanglishBERT, enabling multiple clients to collaboratively train a medical-text language model without sharing raw, sensitive patient data with a central server.

Thesis Students: Faiza Haque Shoily, Muhib Al Ahad, Tanjuma Noor
 Supervisor: Shammi Akhtar
 Co-Supervisor: Md Shafayatul Haque

Scoring Scale: 0 = No valid | 1 = Incorrect | 2 = Partially correct | 3 = Mostly correct | 4 = Completely correct

Please rate each statement below on the 0-4 scale.

The respondent's email (nasimakamal1972@gmail.com) was recorded on submission of this form.

* Indicates required question

Name of the Evaluator *

Nasima Kamal

Occupation *

Doctor

Contact Info (GitHub link, E-mail, LinkedIn, etc.) *

nasimakamal1972@gmail.com

Questionnaire Introduction and Evaluator Details

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 2: Logical Consistency

1. Correct Application of Subject-Specific Concepts

These questions assess whether NLP and Federated Learning concepts, and Bangla/Banglish medical-domain knowledge, are applied correctly in FedMed-BN.

Does the project correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than simply simulating distributed training without true data isolation? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is BanglishBERT an appropriate and correctly used pretrained model for code-mixed Bangla-English medical text? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the FedAvg aggregation mechanism implemented and explained correctly (e.g., weighted averaging by client data size)? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Does the project correctly distinguish next-word prediction (language modelling) from related but different tasks like text classification or NER? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 1: Correct Application of Subject-Specific Concepts

3. Validity and Justification

These questions probe whether design and evaluation decisions are adequately justified with evidence or precedent from the literature.

Is there sufficient justification for choosing BanglishBERT over alternatives such as mBERT or XLM-RoBERTa? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Is the validity of the simulated client partitioning method adequately explained and justified? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Are performance metrics (accuracy, perplexity) appropriate and justified for evaluating next-word prediction specifically? *

0 - No valid 0 1 2 3 4 4 - Completely correct

Section 3: Validity and Justification

Figure D.5: Expert Consultation Evidence - Nasima Kamal (Part 1)

4. Innovation and Contribution

These questions assess the originality and practical value of FedMed-BN relative to existing federated NLP and Bangla medical NLP research.

Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks? *

0

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2

3

4

0 - No valid

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4 - Completely correct

Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource/non-English languages)? *

0

1

2

3

4

0 - No valid

☐

☐

☐

☐

☒

4 - Completely correct

Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context? *

0

1

2

3

4

0 - No valid

☐

☐

☐

☐

☒

4 - Completely correct

Section 4: Innovation and Contribution

5. Overall Quality

These questions ask for a holistic judgment of technical soundness, coherence, and contribution of the thesis as a whole.

Overall, does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem? *

0

1

2

3

4

0 - No valid

☐

☐

☐

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☒

4 - Completely correct

Are the project's stated limitations and future work realistic and proportionate to its current stage? *

0

1

2

3

4

0 - No valid

☐

☐

☐

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☒

4 - Completely correct

Would this work be considered a meaningful and publishable contribution to federated NLP for low-resource medical text? *

0

1

2

3

4

0 - Strongly Disagree

☐

☐

☒

☐

☐

4 - Strongly Agree

Comments or Additional Suggestions *

No comments

Section 5: Overall Quality

Figure D.6: Expert Consultation Evidence - Nasima Kamal (Part 2)

Appendix E

Evidences of CEP

Table E.1: Evidences of CEP

Ps	Attribute	Evidence (Section/Appendix Reference)	Knowledge Profile Mapping (for P1 only)
P1	Depth of Knowledge Required	Chapter 2: Section 2.1 (Preliminaries: Federated Learning, Differential Privacy, Transformer Models, Non-IID Data), Section 2.2 (Related Works), Section 2.3 (Comparative Analysis of Related Works), Section 2.4 (Research Gap / Limitations of Existing Methods). Chapter 3: Section 3.2.1 (High-Level Architecture), Section 3.2.2 (Detailed Design Diagrams), Section 3.2.3 (Design Alternatives and Rationale). Chapter 4: Section 4.2 (Tools, Libraries, and Technologies Used), Section 4.5 (Model Development / Algorithm Design), Section 4.6 (System Implementation / Modules Description).	K3 (Engineering Fundamentals): Chapter 4 - Section 4.2 (Tools, Libraries, and Technologies Used), Section 4.6 (System Implementation). K4 (Specialist Knowledge): Chapter 2 - Section 2.1 (Preliminaries), Chapter 3 - Section 3.2.1 (High-Level Architecture), Section 3.2.3 (Design Alternatives and Rationale). K5 (Engineering Design): Chapter 3 - Section 3.2.1 (High-Level Architecture), Section 3.2.2 (Detailed Design Diagrams). K6 (Engineering Practice): Chapter 4 - Section 4.5 (Model Development), Section 4.6 (System Implementation). K8 (Research Literature): Chapter 2 - Section 2.2 (Related Works), Section 2.3 (Comparative Analysis), Section 2.4 (Research Gap).

Appendix F

Evidences of CEA

Table F.1: Evidences of CEA

As	Attribute	Evidence (Section/Appendix Reference)
A1	Range of Resources	Chapter 2: Section 2.2, 2.3 (Research literature base on FL and privacy-preserving NLP). Chapter 3: Section 3.1 (Stakeholder interviews and questionnaires as human resources). Chapter 4: Section 4.2 (Tools, Libraries, and Technologies Used), Section 4.7 (Hardware and Software Requirements). Chapter 6: Section 6.1 (Project Timeline), Section 6.2 (Budget Estimation).
A2	Level of Interaction	Chapter 3: Section 3.2.2 (Detailed Design Diagrams) - shows how the differential privacy mechanism at each client and FedAvg aggregation at the server interact to balance privacy and model utility; Section 3.2.3 (Design Alternatives and Rationale). Chapter 4: Section 4.5 (Model Development), Section 4.6 (System Implementation).
A3	Innovation	Chapter 3: Section 3.2.3 (Design Alternatives and Rationale) - creative combination of BanglaBERT with DP-FedAvg to enable privacy-preserving collaborative training on Bangla medical text. Chapter 1: Section 1.6 (Expected Outcomes).
A4	Consequences for Society and the Environment	Chapter 1: Section 1.7.1 (Impact on Societal and Cultural Issues), Section 1.7.2 (Impact on Health, Safety, and Legal Issues), Section 1.7.3 (Impact on Environment and Sustainability Issues).
A5	Familiarity	Chapter 1: Section 1.6 (Expected Outcomes) - extends beyond standard coursework into privacy-preserving federated NLP for a low-resource language. Chapter 2: Section 2.4 (Research Gap / Limitations of Existing Methods).